Investing in Your Future? How the Cost of University Shapes Labour Market Outcomes

Luke Walsh*
Department of Economics
University of Liverpool

Abstract

How do students respond when the price of university nearly triples but nothing is paid upfront, and does their response show up in early labour-market outcomes? I study England's 2012 tuition-fee reform using administrative data that link school, university, tax and benefit records for entire cohorts. A difference-in-discontinuities design compares the discontinuity at the September school-entry cutoff for the first affected cohort with the pooled discontinuity for the four preceding cohorts. On-time university entry fell by 4.0 percentage points. Most of this decline was offset by delayed entry: participation within five years was 1.0 points lower and imprecisely estimated, although correcting for the final pre-reform cohort's partial exposure to the reform raises this to a highly significant 2.2 points. Among entrants, attendance at above-mean and top-quartile institutions rose by 4.4 and 3.7 points, more than three times the increases across the whole cohort. Despite the fall in on-time entry, I find no statistically significant effects on employment, out-of-work benefit receipt or annual earnings four to seven years after expected entry, and the estimates rule out employment reductions larger than 1.4 percentage points. A large educational response to price need not translate into detectable changes in early labour-market outcomes.

*Corresponding Author: l.walsh7@liverpool.ac.uk.

1 Introduction

Higher education is a large investment for individuals and for the governments that subsidise it, on the grounds that it raises productivity, earnings and tax revenue. Classic human capital models predict that individuals invest in higher education when the present value of future returns, net of costs, exceeds the value of entering the labour market directly (Becker 1964), so an increase in the price of university should reduce investment at the margin. Two questions follow, and they are rarely answered together. The first is how students respond: whether they enrol, when they enrol, and where. The second is what that response means for the labour-market outcomes of the cohort it affects. A fee increase that deters or delays entry is costly to the extent that the forgone or postponed education would have raised employment and earnings. If early labour-market outcomes do not move, either the education response matters less for early careers than its size suggests, or its consequences lie further ahead.

A widely cited rationale for subsidising higher education is the presence of credit constraints: individuals with strong academic potential but no collateral may be unable to finance tuition even when expected returns are high (Black et al. 2023; Lochner and Monge-Naranjo 2011). England's 2012 tuition fee reform provides a setting in which this friction is largely absent. The reform nearly tripled the cap on annual tuition fees, from £3,375 to £9,000, but all students retained access to government-backed income-contingent loans: no student faced an upfront payment, and repayment remained contingent on future earnings. The fee increase therefore operated through the expected cost of a degree and the debt students would accumulate, not through liquidity.

I study the reform using population-wide administrative data from the Longitudinal Education Outcomes (LEO) dataset, which links school, university, tax and benefit records, so that the same individuals can be followed from secondary school through university entry and into their mid-twenties. The design is a difference-in-discontinuities built on England's 1 September school-entry cutoff. Children born on or after 1 September start school a year later than those

born days earlier, generating a well-documented discontinuity in educational outcomes at the cutoff. Because fees followed the year of university entry, individuals born in September 1993 were the first whose on-time entry fell under the new regime, while those born in August 1993 could enter on time under the old one. I compare the discontinuity at this cutoff with a pooled discontinuity estimated from the four preceding cutoffs, which provides the counterfactual. Month of birth takes only a few values, so I estimate the underlying discontinuities with the framework developed for regression discontinuity designs with a discrete running variable (Kolesár and Rothe 2018; Armstrong and Kolesár 2020), and test the difference with conventional standard errors. For labour-market outcomes, individuals on either side of the cutoff are compared in the same tax year. Because they belong to different school cohorts, they differ in education and experience at that point, so the labour-market discontinuity is not zero even without a reform; the pre-reform discontinuity provides the benchmark against which the 2012 discontinuity is compared.

The paper has three findings. First, the reform substantially reduced immediate university participation. On-time entry fell by 4.02 percentage points, about 16 percent of the on-time entry rate of the last pre-reform cohort. Most of this decline was offset by delayed entry: entry within one year of the on-time date was 1.14 points lower, and entry within five years, by age 23, 1.05 points lower, an estimate that is not significant at the 5 percent level ($p = 0.072$). Whether a meaningful contraction persisted therefore depends on how the final pre-reform cohort is treated. Because fees depended on the year of entry, members of that cohort who delayed entry paid the new fees, which biases the uncorrected estimates for cumulative entry toward zero. Correcting that cohort's outcomes with a model of the entry process of earlier cohorts changes the immediate effect only modestly, to -4.57 points, but doubles the five-year effect, to a highly significant -2.21 points. The immediate response is robust; the persistent response is suggestive in the uncorrected specification and considerably stronger after a correction that rests on modelling assumptions.

Second, the reform changed the composition of university attendance. Conditional on enter-

ing university, students were 4.41 percentage points more likely to attend an above-mean institution and 3.74 points more likely to attend a top-quartile institution. Across the whole cohort, the corresponding increases were 1.23 and 1.14 points. Part of the conditional change reflects the smaller pool of entrants rather than a change in where any given student enrolled, and the gap between the two illustrates why outcomes measured among entrants cannot be read as improvements for the eligible cohort. The conditional response is also present among students below the AAB threshold, and so is not explained by the concurrent exemption of high attainers from student number controls alone.

Third, and in contrast to the education response, I find no statistically significant effects on the principal early labour-market outcomes. Across ages 22 to 25, four to seven years after expected on-time entry, the estimated effects range from -0.13 to -0.46 percentage points for employment, from -0.19 to $+0.66$ points for out-of-work benefit receipt, and from -0.018 to $+0.003$ for log annual earnings (available at ages 22, 24 and 25); none is statistically significant. The confidence intervals rule out reductions in employment larger than about 1.4 percentage points at any age. They do not rule out effects that would be economically meaningful for the comparatively small group whose education the reform changed. The only conventionally significant labour-market estimates are increases in employment in higher-paying industries at age 25, which do not survive a correction for multiple testing and are observed in a tax year that includes the start of the COVID-19 pandemic.

Why does a large education response leave so little trace in early labour-market outcomes? The estimates are reduced-form effects on the whole cohort at the cutoff, and several explanations are consistent with them. The reform changed the education of a minority of the cohort, about 4 percent on time and between 1 and 2 percent within five years, so any effect on those individuals is diluted in cohort-level outcomes. Those who did not enter, or entered later, accumulated labour-market experience earlier, which offsets forgone education in the short run. Students deterred at the margin may have had lower expected returns than the average entrant, and the returns to a

degree may emerge later in working life than the observation window allows. The data cannot separate these explanations. The results therefore do not show that university has no labour-market return, nor that deterred students were better off; they show that a substantial educational response need not produce readily detectable changes in aggregate early labour-market outcomes.

These results contribute to three literatures. The first studies how tuition prices affect participation. Evaluations of the 2012 reform have generally concluded that it had little effect on overall enrolment or on the socioeconomic composition of entrants (Murphy et al. 2019; Azmat and Simion 2021). By isolating the cohort boundary at which the fee regime changed and following each individual's entry for five years, this paper shows a large immediate response, most of which was reversed by delayed entry, and makes explicit how much of the persistent effect depends on the treatment of a partially exposed comparison cohort. The second concerns the labour-market consequences of higher-education finance. Azmat and Simion (2021) report minimal effects of the 2012 reform on early labour-market outcomes; I complement their evidence using linked tax and benefit records for entire cohorts and a design that compares individuals of almost identical age observed in the same tax year, and I set out what the estimates can and cannot rule out. The third concerns where students enrol: conclusions about the destination margin depend on whether outcomes are measured among entrants or across the cohort. Methodologically, the paper shows how contamination of the final pre-reform cohort enters a difference-in-discontinuities design built on adjacent cohorts, and assesses its importance with model-based corrections reported as sensitivity analyses.

The policy implications are specific to England's institutional setting and to the horizon observed. Income-contingent loans removed upfront costs but did not prevent a large immediate fall in university entry. Over the following four to seven years, that fall was not accompanied by detectable deterioration in employment, earnings or out-of-work benefit receipt for the cohort as a whole. This is informative about the short-run costs of the reform, but not about its long-run consequences for earnings, tax revenue or welfare, which depend on returns that accrue later in

working life and that this design does not identify.

The remainder of the paper is structured as follows. Section 2 reviews the related literature. Section 3 provides institutional background on the English higher education system and the 2012 reform. Section 4 describes the data. Section 5 sets out the empirical strategy, including what the discontinuity measures for labour-market outcomes. Section 6 presents the education and labour-market results, Section 7 discusses their interpretation, Section 8 discusses robustness and limitations, and Section 9 concludes. Appendix B describes the correction for the final pre-reform cohort and reports the corrected estimates.

2 Related Literature

Human capital theory predicts that students weigh the costs of university against its expected long-term returns, and that an increase in price should reduce investment at the margin (Becker 1964; Mincer 1974). This prediction concerns the *level* of investment, and empirical work on tuition prices has largely inherited that focus, asking whether fees reduce participation. A price change can also operate on margins that a participation regression cannot see: the timing of entry, particularly when the change is announced in advance, and the choice of institution among those who enter. This paper measures these margins within a single design and links them to early labour-market outcomes for the same cohort.

A large literature examines the participation margin directly. In North America, where tuition varies across states, provinces, and institutions, enrolment falls modestly when sticker prices rise (Kane 1995; Neill 2009; Hemelt and Marcotte 2011); see Deming and Dynarski (2009) for a review. The closest international analogue to the English experience is Germany, where the staggered introduction of tuition fees across states reduced enrolment intentions and transitions into university by small amounts (Hübner 2012; Bruckmeier and Wigger 2014). These studies typically measure enrolment in a fixed window around the price change, and therefore cannot

distinguish delay from a longer-lasting reduction in participation.

A related American literature studies financial aid rather than posted prices, finding that need-based grants raise enrolment and persistence among low-income students (Dynarski 2003; Bettinger 2004; Castleman and Long 2016), with downstream benefits for completion and earnings that can exceed their fiscal cost (Denning et al. 2019; Black et al. 2023). This work is conceptually distinct: aid relaxes liquidity constraints for students who cannot borrow against future earnings, whereas a fee change under a universal income-contingent loan system alters the expected cost of a degree without any liquidity effect. Because no student faces an upfront payment and repayment is contingent on earnings, debt aversion (Callender and Jackson 2005; Field 2009) and responses to the headline price are the main behavioural channels through which such a reform could deter entry.

The UK evidence on tuition fees is mixed. Earlier fee increases reduced participation modestly (Dearden et al. 2011; Sá 2014), and evaluations of the 2012 reform conclude that it had little effect on overall enrolment or on the socioeconomic composition of entrants (Murphy et al. 2019; Azmat and Simion 2021), with Azmat and Simion (2021) additionally reporting minimal effects on early labour market outcomes. Yet the raw entry data show a sharp drop in 2012 (Appendix Figure A.2). My estimates indicate that the drop reflected a substantial causal response, most of which was reversed by delayed entry; whether a meaningful part persisted depends on the treatment of the final pre-reform cohort. Because existing studies rely on application-year snapshots or survey data, they also observe neither entry at later ages nor where a cohort ultimately enrolled; linked administrative data that record entry at any age allow me to measure both, and to follow the same individuals into the labour market.

The labour-market results relate to work on the returns to higher education. In England, Belfield et al. (2018) estimate average earnings returns to attending higher education at age 29 of 26 percent for women and 6 percent for men, and find that men's returns continue to grow after age 30. Returns also vary across individuals: Carneiro et al. (2011) find that returns to

college are higher for those whose unobserved characteristics make them more likely to attend, implying lower returns for students at the margin of enrolment, while Zimmerman (2014) finds that academically marginal students admitted to a four-year college earn returns close to those of the average student. These estimates imply that the cohort-level labour-market effects of a change in participation are likely to be small in early adulthood and to grow with age, which matters for interpreting the estimates here.

The paper also relates to work on pre-announced policies. When agents can shift the timing of a decision, comparisons of outcomes immediately before and after implementation confound the policy's effect with intertemporal substitution around it, and pre-period observations can be contaminated as controls (Malani and Reif 2015). The 2012 reform, announced almost two years before implementation with fees tied to the year of entry, created exactly these conditions. Rather than assume that the final pre-reform cohort provides a clean comparison, I correct its outcomes using the behaviour of earlier cohorts.

Methodologically, the design combines a regression discontinuity with a before–after comparison, a difference-in-discontinuities approach used in other policy settings (Grembi et al. 2016). The running variable, month of birth, is discrete. With a discrete running variable the discontinuity at the cutoff must be extrapolated from a few support points, and the resulting specification error does not vanish in large samples. Lee and Card (2008) proposed clustering standard errors by values of the running variable to account for this error; Kolesár and Rothe (2018) show that such confidence intervals can perform poorly and propose alternatives with guaranteed coverage under explicit restrictions on the conditional expectation function, building on the bias-aware intervals of Armstrong and Kolesár (2020). Section 5.3 explains which parts of this framework the paper uses.

The identifying variation rests on the relative age effect, the well-documented advantage of autumn-born children within academic cohorts (Russell and Startup 1986; McPhillips and Jordan-Black 2009; Crawford et al. 2013, 2014; Black et al. 2011). Season-of-birth selection,

the classic threat to birthdate designs (Buckles and Hungerman 2013), is differenced out to the extent that it is stable across cohorts, and I report balance tests on predetermined characteristics for each cohort in the sample.

3 Institutional Background

In the decades following WWII, the funding of students into higher education was largely administered by local authorities, an approach formalised by the Education Act (1962). The Act mandated that Local Education Authorities pay tuition fees and provide means-tested maintenance grants for eligible full-time university students (UK Parliament 1962), making undergraduate education effectively free at the point of use across the UK. The economic rationale was summarised in the Robbins Report of 1963, which recommended a substantial expansion of the university system on the grounds that higher education generated considerable positive social and financial externalities, such that growth in the economy and in tax receipts would deliver a positive return on public investment (Committee on Higher Education 1963).

This assumption came under strain in the 1990s. Student numbers had grown rapidly while funding per pupil declined by 40% between the early 1980s and 1997, putting considerable pressure on university budgets (Dearden et al. 2011). At the same time, the case for higher education paying for itself at the exchequer level had begun to wane: the Dearing Report (1997) argued that a cost-sharing approach would be equitable since graduates were the primary beneficiaries of university education (National Committee of Inquiry into Higher Education 1997). Subsequent evidence supports this reassessment: Britton et al. (2020) estimate that while average lifetime returns to the exchequer are positive net of subsidies, these returns are driven predominantly by the highest earners, and total returns are negative for up to 30% of graduates.

3.1 Post-1998 Reforms

Following the Dearing Report, the Teaching and Higher Education Act (1998) introduced annual tuition fees of £1,000, paid upfront by students, while maintenance grants were replaced with income-contingent maintenance loans (UK-Government 1998; Hillman 2013). Exemptions applied to students from low-income families, and following devolution in 1999 the UK nations began to diverge: Scotland introduced a graduate endowment in lieu of upfront fees and by 2007 had abolished tuition fees entirely for Scottish-domiciled undergraduates at Scottish institutions (Sá 2014).

By the early 2000s the £1,000 cap was yielding limited revenue, and the requirement that fees be paid upfront was widely seen as a barrier to access. The Higher Education Act 2004 raised the cap to £3,000 per year from the 2006–07 academic year and, crucially, ended upfront payment altogether (UK Parliament 2004). Fees would instead be covered by government-backed loans, repaid after graduation at 9% of earnings above a £15,000 threshold, with any outstanding balance written off after 25 years. University thereby became free at the point of use once more, with higher fees operating as a reduction in expected lifetime income rather than a liquidity burden. This institutional detail distinguishes the UK setting from the US, where the policy debate centres on the under-allocation of capital to students who cannot collateralise their future earnings, and it is what allows this paper to interpret responses to the 2012 reform as responses to price rather than to credit constraints.

3.2 The 2012 Reform and Its Announcement

By 2010, recession and the declining real value of the capped fee had renewed the pressure on university finances. The independent Browne Review, published in October 2010 recommended a market-driven abolition of the fee cap, effective for students entering university from the 2012–13 academic year (Independent Review of Higher Education Funding and Student Finance 2010).

This was accepted by the government in November, but with the price cap rising to £9,000 rather than being abolished. In theory variable fees were intended to generate price competition; in practice almost all universities charged the maximum immediately upon implementation (House of Commons Library 2018). The reform also shifted the repayment threshold from just below £16,000 to £21,000, extended the repayment period from 25 to 30 years, and introduced a higher, income-contingent interest rate (House of Commons Library 2018). The net effect was a more progressive schedule: the lowest-earning graduates could expect to pay less per month than under the previous system (Britton et al. 2017). Appendix Figure A.1 summarises the evolution of the fee cap, with the 2012 increase standing as a clear outlier against the earlier reforms.

Two features of the reform's implementation are central to the empirical design. First, the fee regime a student faced was determined by their year of entry, not their year of birth: anyone commencing study before September 2012 paid the old fees for the duration of their course, while anyone commencing from September 2012 onwards paid the new fees. Second, the reform was announced almost two years before it took effect. Together these features created incentives to retime entry around the reform. Students in the final pre-reform cohort could lock in the lower fees by entering in 2011 rather than deferring, but those who nevertheless deferred entered under the new fees; students in the first treated cohort faced the higher fees whenever they entered. Appendix Figure A.4 shows that the share of students delaying entry by one year fell in 2011 by just over two percentage points. The final pre-reform cohort was therefore partially exposed to the reform, which Section 5.6 shows has important consequences for the estimates and corrects.

Appendix Figure A.2 displays the raw trend in domestic undergraduate enrolment over the sample period: steady year-on-year growth, a sharp contraction in 2012, and a recovery in subsequent years. Taken at face value, the 2012 dip suggests a substantial deterrent effect of the fee increase, but raw entry counts mix the response to the reform with trends in cohort size and participation and with shifts in the timing of entry. The task of this paper is to estimate how much of the change reflects a causal response to the fee increase, how much of that response was

postponement, and how the reform changed where students enrolled.

3.3 Concurrent Policies

Three contemporaneous policy changes could in principle confound the analysis. First, and most importantly, the regulation of student numbers changed in 2012–13. Entry controls had limited the number of fundable full-time undergraduate entrants at each institution since 2009–10. For 2012–13 entry, students with qualifications equivalent to AAB or higher at A-level were removed from these controls, and up to 20,000 places were withdrawn from institutions' core allocations and made available to institutions charging an average net fee of £7,500 or less (Higher Education Funding Council for England 2011). The exemption was extended to ABB+ in 2013–14, and number controls were abolished entirely from 2015–16. The AAB exemption gave selective institutions freedom to expand recruitment of high-attaining students at the moment the fee reform took effect, and is therefore a candidate explanation for any shift in enrolment toward selective institutions. Section 6.3 examines whether the shift is present among students below the AAB threshold. Second, the participation age was raised, requiring young people in England to remain in education or training until age 17 from 2013 and 18 from 2015; this affects younger cohorts than those studied here. Third, maintenance support evolved over the period, but these changes were incremental and applied smoothly across cohorts (House of Commons Library 2018), so that they are differenced out to the extent that they did not change discontinuously at the 2012 cutoff.

4 Data and Descriptive Statistics

This paper uses data from the Longitudinal Education Outcomes (LEO) dataset. This administrative panel links school census records from the National Pupil Database (NPD) on student characteristics to Higher Education Statistics Agency (HESA) records of university enrolment

and outcomes, matched further to HM Revenue and Customs (HMRC) records of employment and earnings and to Department for Work and Pensions (DWP) records of benefit receipt. The result is a longitudinal panel covering over 30 million people across a 23-year period (1998–2020), from which I construct cohorts of English-domiciled, state-educated pupils defined by academic year of birth. Because the panel follows individuals rather than entry classes, I observe university entry at any age, not merely entry in the year following a student's 18th birthday, and I observe the same individuals in the labour market into their mid-twenties.

Enrolment. For each individual I construct an indicator for on-time entry, defined as entry in the academic year following the individual's 18th birthday, and indicators for cumulative entry within one to five years of that date. Entry within five years corresponds to entry by age 23, and for the 2012 cohort requires HESA coverage through the 2017–18 academic year, which the panel provides.

Selectivity. Institution selectivity is measured by ranking institutions according to the mean A-level points of their entering students in the 2011 pre-reform reference year, from which I define indicators for attending an institution above the mean, in the top quartile, in the top 20 percent, and in the top 10 percent of the distribution of percentiles of institutions. Each indicator is constructed in two forms. The *conditional* form is defined among individuals observed entering university and describes where entrants enrolled. The *unconditional* form is defined for the whole cohort and equals zero for individuals who do not enter, so that it measures the share of the cohort attending an institution of a given selectivity.

Subject choice, completion and progression. I also measure whether a student enrols in a STEM subject (conditionally and unconditionally), completes the degree, progresses to postgraduate study, or withdraws because of poor academic performance. The withdrawal indicator is constructed from the HESA reason for ending a course and isolates withdrawal for poor aca-

demic performance; it is not a general measure of non-completion. Conditional samples differ slightly across these outcomes.

Labour-market outcomes. Labour-market outcomes are measured in the tax years four to seven years after the expected on-time university entry year of individuals born immediately after each cutoff, corresponding approximately to ages 22 to 25. Individuals on both sides of a cutoff are observed in the same tax year (Section 5.5). The outcomes are an indicator for being employed and log annual earnings in 2015 prices, both drawn from HMRC Pay As You Earn (PAYE) records; an indicator for receiving out-of-work benefits; and indicators for being employed in an industry whose mean pay is above the mean, or in the top quartile, of industries ranked by mean pay in a cross-section of the 2015 tax year.[1] The estimation samples for employment, out-of-work benefit receipt and the two industry indicators are the full cohort in each window, so these outcomes are defined for the whole cohort rather than only among the employed. Earnings are defined only for individuals with recorded earnings, between 75 and 81 percent of each window, so effects on earnings combine any change in pay with any change in who has recorded earnings. The final tax year in the data, 2019–20, provides the age-25 observation for the 2012 cohort and includes the start of the COVID-19 pandemic.

Characteristics. Personal characteristics recorded in the NPD are used for balance tests and the correction described in Section 5.6. Family income is not recorded, so eligibility for free school meals (FSM) in year 11, a means-tested programme for children from low-income families, is used as a proxy for childhood poverty. Neighbourhood disadvantage is measured by school residence in a local area with low university participation (LOPAR). Prior attainment is measured by the number of GCSEs achieved at each grade, and by A-level results, from which I construct an indicator for achieving grades of AAB or higher, the threshold for the 2012–13 exemption

[1] LEO does not contain occupation codes, so industry allocation may not map one-for-one into occupational upgrading. The ranking of industries is highly robust to the choice of reference year.

from student number controls (Section 3.3). I also observe each student's school and Lower Layer Super Output Area (LSOA) of residence.[2]

Table 1 presents summary statistics for first-year university entrants immediately before and after the reform. The composition of entrants remained broadly stable between the 2011 and 2012 cohorts: the share of female students was unchanged, and the shares eligible for FSM and from low-participation areas rose marginally. Two margins moved. Entry in the year following students' 18th birthdays fell by 2.3 percentage points, and the share of entrants attending an above-average institution rose from 39 to 43 percent. Neither difference can be interpreted causally: raw comparisons of adjacent cohorts mix the reform with trends in participation and with the partial exposure of the 2011 cohort, which the design in Section 5 addresses.

Table 1: University Entrant Characteristics Before and After the 2012 Fee Reform

	2011	**2012**
Entry into university at 18 (%)	24.8	22.5
Female (%)	55.7	55.7
Eligible for FSM (%)	9.0	9.6
From low university-share area (%)	56.0	56.8
AAB or higher at A-level (%)	20.2	20.3
Attending above-average institution (%)	39	43

Notes: This table reports the share of first-year university entrants by key demographic and background characteristics in 2011 (pre-reform) and 2012 (post-reform). FSM = Free School Meals. The first row shows entry rate with the entire cohort as the denominator, while subsequent rows are conditional on entering HE. "Low university-share area" indicates local areas with below-average university participation in 2011. "Attending above-average institution" refers to institutions ranked above the mean in terms of average A-level points of entrants. Counts rounded to the nearest five per LEO disclosure rules.

5 Empirical Strategy

This paper asks whether a sharp increase in the cost of university affected whether, when and where students attended, and whether those responses changed early labour-market outcomes.

[2]In accordance with the disclosure rules governing LEO, all reported counts are rounded to the nearest five.

Estimating these effects is difficult: tuition fees typically change alongside other policies, participation was growing over the period, macroeconomic conditions shift the value of study relative to work, and, in this case, the reform's announcement changed the behaviour of the cohort immediately before it. A simple comparison of adjacent cohorts cannot separate the fee increase from these influences.

5.1 The School-Entry Discontinuity

In England, children are placed into academic cohorts by date of birth, with a cutoff on 1 September. Those born just before the cutoff (in August) enter school a full year earlier than those born just after (in September), even though they are only days apart in age. This generates sharp and well-documented differences in long-term outcomes, including university attendance, between children born on either side of the cutoff, commonly referred to as the relative age effect (Russell and Startup 1986; McPhillips and Jordan-Black 2009; Campbell 2013; Crawford et al. 2013, 2014). Appendix Figure A.3 shows this premium in university enrolment: September-born individuals are around four percentage points more likely to enrol than those born in August, with enrolment declining across subsequent birth months.

I index each cutoff by t, the academic year in which individuals born immediately after it would enter university on time. Individuals born in the months before cutoff t belong to entry cohort $t-1$ and are the youngest in their school year; those born in the months after it belong to entry cohort t and are the oldest in theirs. The reform cutoff is $t = 2012$: individuals born in September 1993 were the first whose on-time entry fell under the new fees, while those born in August 1993 belong to the 2011 entry cohort and could enter on time under the old regime. The pre-reform cutoffs are $t \in \{2008, 2009, 2010, 2011\}$.

Let x_i denote individual i's month of birth measured relative to the cutoff, and Y_i an outcome.

The discontinuity at cutoff t is

$$\tau_t \;=\; \lim_{x \downarrow 0} \mathbb{E}\left[Y_i \mid x_i = x,\, t\right] \;-\; \lim_{x \uparrow 0} \mathbb{E}\left[Y_i \mid x_i = x,\, t\right]. \tag{1}$$

Because month of birth is discrete, the conditional expectation function is observed only at a small number of support points, and the limits in (1) are extrapolations from the points nearest the cutoff. Section 5.3 returns to the implications.

5.2 The Difference-in-Discontinuities Estimator

In every cohort, τ_t reflects the relative age effect together with any differences between the adjacent cohorts on either side of the cutoff. The pre-reform discontinuity therefore provides the counterfactual for 2012: had the reform not occurred, the discontinuity at the 2012 cutoff should have resembled those at earlier cutoffs. The main estimate is

$$\widehat{\Delta} \;=\; \widehat{\tau}_{2012} \;-\; \widehat{\tau}_{2008\text{–}2011}, \tag{2}$$

where $\widehat{\tau}_{2012}$ is the estimated discontinuity at the 2012 cutoff and $\widehat{\tau}_{2008\text{–}2011}$ is estimated from a single regression discontinuity fitted to the pooled observations from the four pre-reform cutoffs, with the running variable measured relative to each year's cutoff alongside cohort fixed effects. The pooled estimate uses all pre-reform observations in one fit; it need not equal the arithmetic average of four separately estimated annual discontinuities.

Let θ denote the effect of the reform on individuals at the 2012 cutoff, and τ^0_{2012} the discontinuity that would have been observed at that cutoff without the reform. $\widehat{\Delta}$ estimates θ if $\tau^0_{2012} = \tau_{2008\text{–}2011}$: absent the reform, the discontinuity at the 2012 cutoff would have equalled the pooled pre-reform discontinuity.

To see what this assumption requires, it is useful to decompose each discontinuity. Suppose,

for exposition, that the expected outcome of an individual in entry cohort k with relative birth position x can be written as $m(x) + \mu_k$, where m is a relative-age profile, common to all cohorts, that jumps by ρ at the cutoff, and μ_k is a shift common to all members of cohort k. Then $\tau_t = \rho + \mu_t - \mu_{t-1}$. If the pooled estimator places weight ω_t on cutoff t, with $\sum_t \omega_t = 1$, the estimand of (2) is

$$\Delta \;=\; \theta \;+\; (\mu_{2012} - \mu_{2011}) \;-\; \sum_{t=2008}^{2011} \omega_t \,(\mu_t - \mu_{t-1})\,. \tag{3}$$

Three implications follow. First, the relative age effect ρ cancels, as does any season-of-birth selection that is stable across cohorts. Second, when the weights are approximately equal, cohort-level shifts that follow a linear trend also cancel, because the change between the 2011 and 2012 cohorts then equals the average annual change over the pre-reform period; the design therefore accommodates the secular growth in participation over this period. Third, a shift specific to the 2011 cohort enters Δ with coefficient $-(1 + \omega_{2011})$, because that cohort supplies both the comparison group immediately below the 2012 cutoff and the group immediately above the 2011 cutoff. Any effect of the reform on the 2011 cohort is therefore transmitted more than one-for-one into the estimate. Section 5.6 addresses this.

Inference on Δ uses the standard errors of the two estimated discontinuities and assumes that their covariance is zero:

$$SE\big(\widehat{\Delta}\big) \;=\; \sqrt{SE\big(\widehat{\tau}_{2012}\big)^2 \;+\; SE\big(\widehat{\tau}_{2008\text{–}2011}\big)^2}\,. \tag{4}$$

I test $H_0 : \Delta = 0$ with the conventional two-sided test based on $\widehat{\Delta}/SE(\widehat{\Delta})$ and standard normal critical values. All p-values and significance levels reported for the main estimates are from this test. The zero-covariance assumption holds exactly under independent sampling of individuals, because the estimation windows described below ensure that no individual contributes to both $\widehat{\tau}_{2012}$ and $\widehat{\tau}_{2008\text{–}2011}$.

5.3 Estimation with a Discrete Running Variable

Month of birth takes only twelve distinct values in each estimation window, six on each side of the cutoff. With a running variable of this kind, the discontinuity at the cutoff must be extrapolated from a few support points. The bandwidth cannot shrink below the distance between support points, so the bias of a local polynomial estimator, its specification error, does not vanish as the sample grows, and inference that ignores it can be misleading however large the sample. A common response has been to cluster standard errors by values of the running variable (Lee and Card 2008). Kolesár and Rothe (2018) show that such confidence intervals do not guard against misspecification and can have poor coverage, and propose alternatives with guaranteed coverage under explicit restrictions on the conditional expectation function, closely related to the bias-aware construction of Armstrong and Kolesár (2020), in which the critical value is inflated to account for the largest bias consistent with a bound on the curvature of the conditional expectation function.

I estimate each discontinuity by local linear regression within a fixed window, using the RD-Honest implementation of these methods using a triangular kernel, which assigns greater weight to observations closer to the September school-entry cutoff. It is important to be precise about which elements of this framework the paper uses. Because the window is fixed by the researcher rather than chosen by the procedure, the point estimates do not depend on the smoothness bound that the bias-aware procedure uses to select bandwidths and critical values: they are local linear regression estimates. The standard errors are the heteroskedasticity-robust standard errors produced by the procedure and are not clustered by month of birth, following the recommendation of Kolesár and Rothe (2018). Standard errors for the underlying regression discontinuity estimates are calculated using the Eicker–Huber–White heteroskedasticity-robust variance estimator, without clustering. The bias-aware critical values are not used for the main tests. The inference reported in this paper is therefore conventional inference applied to RDHonest point estimates and standard errors, not bias-aware inference, and it inherits the limitations set out in Section 5.4.

Two windows. The pooled estimator in (2) uses a 12-month window, six birth months on each side of each cutoff. With this window, the observations around cutoff t are the younger half of entry cohort $t-1$ and the older half of cohort t, so the windows around adjacent cutoffs are disjoint and no individual contributes to more than one discontinuity. A wider window would place the same individuals above one cutoff and below the next, so that the pooled pre-reform estimate would use some individuals twice and the two terms in (2) would share observations. The annual figures use a 24-month window, twelve months on each side, to display the evolution of the discontinuity across cohorts using the full cohorts on either side of each cutoff. Adjacent annual estimates in the figures therefore share an entire cohort and are not independent, and because they use a wider window than the pooled estimator, differences between plotted estimates need not reproduce the pooled estimates. The annual estimates are supporting visual and diagnostic evidence that simultaneously serve as a bandwidth check, since the formal test only has 12 mass points around the cutoff; formal estimates and tests come from (2).

5.4 Inference

The conventional test in Section 5.2 abstracts from two sources of error.

Smoothing bias. The test treats each discontinuity estimate as unbiased. With a discrete running variable this is an assumption, not an approximation that improves with sample size. Bias-aware confidence intervals of the kind proposed by Armstrong and Kolesár (2020) and Kolesár and Rothe (2018) avoid the assumption, but with six support points on each side of the cutoff the worst-case bias they must accommodate is large relative to sampling error, and bounding the bias of each discontinuity separately compounds the problem for a difference between two discontinuities. Such intervals can therefore be much wider than conventional ones. The difference-in-discontinuities design offers a reason to expect the relevant bias to be smaller than these worst-case bounds: the bias of each local linear estimate depends on the curvature of the

conditional expectation function near the cutoff, and to the extent that the relative-age profile has a similar shape in the 2012 and pre-reform cohorts, the biases of the two terms in (2) are similar and difference out. This is a maintained assumption, not a result. The reported p-values are valid under it, and do not carry the coverage guarantees of bias-aware procedures.

Cohort-level shocks. The standard errors reflect sampling variation across individuals within each window, not shocks common to an entire cohort. Equation (3) shows that cohort-level shifts that do not follow a trend enter Δ directly. Because the 2011 cohort contributes to both terms of (2), a shock to that cohort moves them in opposite directions; the zero-covariance assumption is exact under independent sampling of individuals but not in the presence of such shocks. With a single treated cohort, the variance of cohort-level shocks cannot be estimated from the treated side, and cluster-robust and wild-bootstrap procedures are unreliable with one treated cluster (Cameron et al. 2008; MacKinnon and Webb 2018; Conley and Taber 2011). The annual estimates provide an informal check: if the pre-reform discontinuities vary across cohorts by more than their standard errors imply, cohort-level shocks are material.

5.5 Labour-Market Outcomes: What the Discontinuity Measures

The same estimator is applied to labour-market outcomes, but the discontinuity it measures has a different interpretation. For each cutoff t and follow-up horizon $k \in \{4, 5, 6, 7\}$, individuals on both sides of the cutoff are observed in the same tax year, k years after the on-time entry year of those born just after the cutoff. Because the two groups are only days apart in age and are observed at the same date, macroeconomic conditions and the calendar-time environment are held approximately constant across the cutoff. The two groups belong, however, to different school cohorts. Those born just before the cutoff started school, completed compulsory schooling, and could have entered and completed university a year earlier than those born just after it, so at a given date they have, on average, progressed a year further through education and had up to a

year more of potential labour-market experience.

Three components of the labour-market discontinuity should be distinguished. The first is the cross-sectional difference in education and experience within a tax year, which arises in every cohort. It implies that the discontinuity $\tau_{t,k}$ is generally not zero, even without a reform. In the pre-reform cohorts, the pooled discontinuity in log annual earnings is -0.197 at age 22, -0.069 at age 24 and -0.039 at age 25 (Table 3): individuals born just after the cutoff earn substantially less at 22, when many are still in, or have only just left, education while their slightly older counterparts are already in work, and the gap narrows as both groups settle into the labour market. The second is variation in this experience-related discontinuity across cohorts, which could arise if, for example, the value of an additional year of experience depended on labour-market conditions when each cohort entered work. The third is any change in the discontinuity caused by the reform. The difference-in-discontinuities estimand,

$$\Delta_k \;=\; \tau_{2012,k} \;-\; \tau_{2008\text{–}2011,k}, \tag{5}$$

removes the first component and identifies the third under the assumption that the second is absent: without the reform, the discontinuity associated with school-cohort membership and experience at horizon k would have been the same at the 2012 cutoff as at the pooled pre-reform cutoffs. The identifying restriction is therefore the stability of the counterfactual discontinuity across cohorts, not the absence of an experience-related discontinuity. Each horizon is estimated separately: the pre-reform discontinuity at horizon k pools the four pre-reform cutoffs at that horizon, not observations across horizons.

The estimates are consistent with this assumption, although they cannot test it. The underlying discontinuities are large relative to the differences between them: at age 22, the 2012 earnings discontinuity is -0.215 against -0.197 before the reform, and at age 25, -0.036 against -0.039. The annual discontinuities (not reported) are of similar magnitude in each pre-reform cohort and

in 2012, consistent with a stable experience-related discontinuity and no large additional change in 2012.

Two further issues are specific to these outcomes. First, earnings are observed only for individuals with recorded earnings, so the earnings estimates combine changes in pay with changes in selection into having earnings, which the reform could itself affect. Second, the age-25 observation for the 2012 cohort falls in the 2019–20 tax year, which includes the start of the COVID-19 pandemic and the first national lockdown in March 2020, whereas the corresponding pre-reform observations predate the pandemic. If the pandemic affected individuals in adjacent school cohorts differently, for example because they were at different stages of education or early careers, it could have shifted the 2012 discontinuity at age 25 for reasons unrelated to the reform. The direction of any such effect is not known. I therefore concentrate on ages 22 to 24 and report age 25 for completeness.

5.6 Contamination of the Final Pre-Reform Cohort

The problem. The fee a student paid depended on the year in which they entered university. Members of the 2011 entry cohort who delayed entry by a year, for example after a gap year, entered in 2012 under the new fees, as did members of the 2010 cohort who delayed by two years. Because the reform was announced in late 2010, members of the 2011 cohort could also respond in anticipation, either by entering on time rather than deferring or by abandoning plans for delayed entry. The nominally untreated 2011 cohort was therefore partially treated, and so the pre-reform counterfactual is contaminated.

Equation (3) shows why this matters. Suppose the higher fees deterred some of the 2011 cohort's would-be delayed entrants, lowering that cohort's cumulative entry by $\delta > 0$. This lowers μ_{2011} by δ and so raises Δ by $(1 + \omega_{2011})\delta$, biasing the estimated effect on cumulative entry toward zero. Most of this attenuation operates through the comparison side of the 2012 cutoff, because the individuals immediately below that cutoff belong to the 2011 cohort; the

remainder operates through the pre-reform benchmark. Without a correction, the estimand for cumulative outcomes is therefore not the effect of the new fee regime relative to the old one. It is the effect of losing access to the old fee for on-time entry, relative to a group that retained it for on-time entry but faced the new fees if it delayed.

I assess the importance of this contamination with a correction applied to the whole 2011 entry cohort (individuals born between September 1992 and August 1993), who appear above the 2011 cutoff in the pre-reform pool and below the 2012 cutoff on the comparison side of the treated discontinuity. For outcomes defined for the whole cohort, the correction replaces the 2011 cohort's outcomes with predictions from models estimated on the 2008–2010 cohorts. For outcomes defined among university entrants, it reweights the 2011 cohort's entrants towards the composition that the earlier selection process would have produced. For labour-market outcomes, it reweights the 2011 cohort using predicted probabilities of on-time university entry. Each version rests on modelling assumptions, most importantly that the 2011 cohort would otherwise have followed the earlier cohorts' process extended by a linear trend, and the standard errors of the corrected estimates do not reflect estimation of the models. The labour-market version is a weaker correction still: the characteristics that predict on-time entry need not be those that determine employment, earnings or industry, and selection into employment and into having earnings could itself respond to the reform. Appendix B describes the procedures, their assumptions and the remaining contamination of earlier cohorts, and reports all corrected estimates. Throughout, I treat the uncorrected estimates as the primary specification and the corrected estimates as sensitivity analyses. Contamination operates through delayed entry, so it matters most for cumulative entry, where the uncorrected estimates are likely to be biased toward zero.

5.7 Threats to Identification and Covariate Balance

The identifying assumption is that, apart from the reform, nothing changed the discontinuity at the 2012 cutoff relative to the pre-reform discontinuity. Equation (3) makes clear that the threat

is a shock specific to the 2011 or 2012 cohort that departs from the pre-reform trend: concurrent policies, changes in the composition of births around the cutoff, or changes in the timing of school entry.

Covariate balance. Appendix Table A.1 reports regression discontinuity estimates for three predetermined characteristics at each cutoff from 2008 to 2012. The results are mixed. There is no statistically significant discontinuity in the share female in any year. For FSM eligibility, the reported confidence intervals exclude zero in 2008 (-2.9 percentage points) and 2010 (-1.1), and for residence in a low-participation area they exclude zero in 2011 (-1.5). None of the three discontinuities is significant at the 2012 cutoff. Balance therefore does not hold in every pre-reform cohort. One possible reason is that characteristics such as FSM eligibility are recorded at a fixed school stage, so the two sides of each cutoff are measured a year apart, and changes in eligibility or recording between consecutive years appear as discontinuities.

Individually significant pre-reform discontinuities do not by themselves invalidate the design, which requires that the discontinuity at the 2012 cutoff equal the pooled pre-reform discontinuity in the absence of the reform, not that every annual discontinuity be zero. What matters is whether the covariate discontinuities differ between 2012 and the pre-reform period. Relative to the average of the four annual pre-reform estimates, the 2012 discontinuity is about 1.2 percentage points higher for FSM eligibility, 1.0 points higher for low-participation residence, and 0.5 points lower for the share female. To gauge whether differences of this size matter, I multiply each by the raw difference in the outcome between the corresponding groups in the 2012 estimation window. For five-year entry, the implied compositional contributions sum to about -0.3 percentage points, and for on-time entry to about -0.2 points, in both cases in the direction of a larger estimated contraction; for unconditional top-quartile attendance and STEM participation they are about -0.1 points. Because FSM eligibility and low-participation residence are correlated, summing their contributions overstates the combined effect, and the calculation uses rounded estimates; it

indicates only that compositional differences of the observed size could account for a modest part of the enrolment estimates and are unlikely to account for their sign or most of their magnitude

Strategic delay of school entry. Some summer-born children defer school entry to avoid the relative age effect, a practice concentrated among higher-income families. For this to bias the results, there would need to be a cohort-specific change in delayed school entry for children who would later face the 2012 fees, that is, among children starting school around 1998. Appendix Figure A.6 plots the share of students delaying primary school entry by university-entry cohort: the proportion is stable at approximately 0.5 percent, with no discontinuity around the treated cohort. The main analysis excludes individuals whose school-entry timing does not align with their birthdate cohort

Seasonality of birth. Family background varies with season of birth (Buckles and Hungerman 2013), and the relative age premium partly reflects this selection. The design does not require births to be random across months, only that the selection pattern be stable across cohorts, in which case it is differenced out; the balance evidence above bears on this.

Concurrent policies. The raising of the participation age affects later cohorts, and changes to maintenance support were incremental (Section 3.3). The AAB exemption from number controls applied to exactly the cohort treated by the fee increase and is examined directly in Section 6.3.

6 Results

All estimates are pooled difference-in-discontinuities estimates from (2), in the uncorrected specification unless stated otherwise. Significance refers to the conventional test described in Section 5.2, subject to the caveats in Section 5.4. Sections 6.1 to 6.4 report education outcomes and Section 6.5 reports labour-market outcomes.

Table 2: The Effect of the Reform on University Entry and Institution Selectivity

	(1) Estimate	(2) N, pre-reform	(3) N, 2012
Panel A. University entry (whole cohort)			
On-time entry (age 18)	-4.02^{***}	986,270	242,865
	[<0.001]		
Entered within 1 year (by age 19)	-1.14^{**}	986,275	242,865
	[0.040]		
Entered within 5 years (by age 23)	-1.05^{*}	986,270	242,865
	[0.072]		
Panel B. Institution selectivity among university entrants			
Above-mean institution	4.41^{***}	345,055	87,005
	[<0.001]		
Top-quartile institution	3.74^{***}	345,055	87,005
	[<0.001]		
Panel C. Institution selectivity across the whole cohort			
Above-mean institution	1.23^{***}	986,275	242,865
	[0.004]		
Top-quartile institution	1.14^{***}	986,275	242,865
	[<0.001]		

Notes: Column (1) reports $\widehat{\Delta} = \widehat{\tau}_{2012} - \widehat{\tau}_{2008\text{–}2011}$ in percentage points (equation 2) in the uncorrected specification. Each discontinuity is a local linear RDHonest estimate with a triangular kernel within a 12-month window (six birth months on each side of the 1 September cutoff); the pre-reform discontinuity is estimated on the pooled observations from the 2008–2011 cutoffs. P-values in brackets are from a conventional two-sided test using $SE(\widehat{\Delta}) = [SE(\widehat{\tau}_{2012})^2 + SE(\widehat{\tau}_{2008\text{–}2011})^2]^{1/2}$ with Eicker–Huber–White standard errors; they are not bias-aware. Institutions are ranked by the mean A-level points of their entrants in 2011. Panel B is defined among university entrants; Panel C is defined for the whole cohort and equals zero for individuals who do not enter. Intermediate horizons, further selectivity thresholds and corrected estimates are in Appendix Tables B.1 and B.2. Counts rounded to the nearest five. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

6.1 University Participation: Immediate Response, Delay, and Persistence

Panel A of Table 2 and Figure 1 report the effects on university entry. On-time entry fell by 4.02 percentage points ($p < 0.001$), about 16 percent of the 24.8 percent on-time entry rate of the 2011 cohort (Table 1). Entry within one year of the on-time date was 1.14 points lower ($p = 0.040$), so around 70 percent of the immediate decline was reversed by students entering a year late. Entry within five years, by age 23, was 1.05 points lower ($p = 0.072$), and the estimates at intermediate horizons are similar (-1.15, -1.16 and -1.08 points at two, three and four years; Appendix Table B.1 and Appendix Figure A.7). In the primary specification, then, most of the immediate response was postponement, and the remaining contraction of about one percentage point is not statistically significant at the 5 percent level.

The persistence of the contraction depends more heavily on the treatment of the 2011 cohort than the immediate response does. Correcting that cohort's outcomes increases the on-time estimate only modestly, to -4.57 points, but doubles the five-year estimate, to -2.21 points ($p < 0.001$; Appendix Table B.1, and panels (b) and (d) of Figure 1). This is the pattern expected if the uncorrected cumulative estimates are attenuated by the 2011 cohort's treated delayed entry (Section 5.6): that contamination first affects cumulative outcomes at the one-year horizon, which is where the gap between the two specifications opens. The corrected estimates, however, depend on extrapolating the entry process of earlier cohorts. The immediate fall in entry is therefore large and robust. The evidence points to a persistent contraction of between about one and two percentage points, suggestive in the uncorrected specification and considerably stronger after correction, but neither estimate should be read as definitive evidence of permanent non-participation.

The dynamic estimates require one further caution. The exposure of the pre-reform cohorts to the new fees changes with the horizon: within one year, only the 2011 cohort could have entered under the new fees, while at horizons of two to five years, members of earlier cohorts who delayed for longer could also have done so. Exposure that is spread evenly within a cohort largely cancels in the pooled pre-reform estimate (Appendix B), but differences between horizons

in the composition of exposure limit how finely the dynamic path can be interpreted. I therefore focus on the immediate effect, the one-year horizon, which captures most delayed entry, and the five-year horizon.

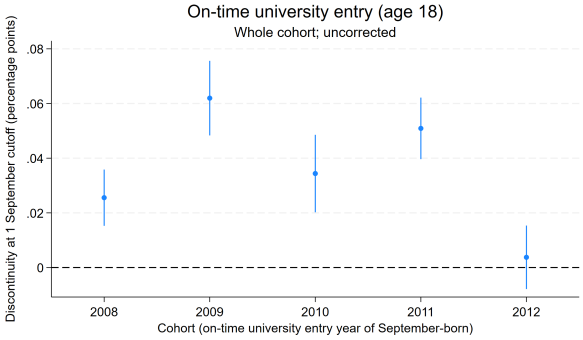
The distinction between the on-time and five-year estimates is between postponing entry and not having entered by age 23. Postponement is consistent with students re-optimising the timing of entry, working or taking time out before committing to a more expensive degree; not having entered by 23 is consistent with the fee increase moving some marginal students out of higher education over the horizon observed. Neither estimate establishes permanent non-participation: some of those who had not entered by 23 may enter later, for example as mature students.

6.2 The Composition of University Participation

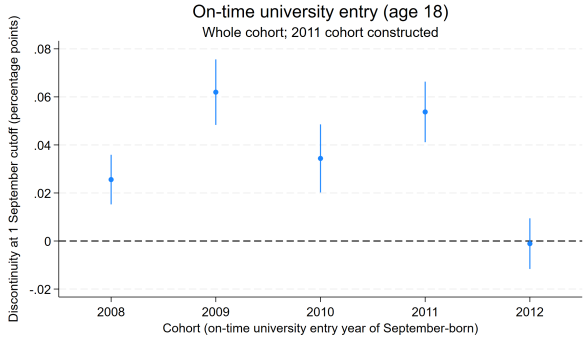
Panels B and C of Table 2 and Figure 2 report effects on the selectivity of the institution attended. Among university entrants, the probability of attending an above-mean institution rose by 4.41 percentage points and that of attending a top-quartile institution by 3.74 points (both $p < 0.001$), with smaller increases at the top 20 percent (2.50 points) and top 10 percent (1.47 points) of institutions (Appendix Table B.2). Relative to the shares of entrants attending such institutions in the 2012 estimation window, about 39 and 21 percent, these are increases of roughly 11 and 18 percent. Descriptively, the share of entrants attending higher-ranked institutions jumps sharply after 2012 (Appendix Figure A.5).

Across the whole cohort, the corresponding effects are 1.23 points for above-mean institutions ($p = 0.004$) and 1.14 points for top-quartile institutions ($p < 0.001$), less than a third of the conditional estimates. The two sets of estimates are linked mechanically: the conditional share is the unconditional share divided by the entry rate, so a fall in entry raises the share of entrants at selective institutions even if no individual changes destination. In the 2012 estimation window, about 36 percent of the cohort entered university within five years, and about 39 percent of entrants attended an above-mean institution. A fall of 1.05 points in participation, concen-

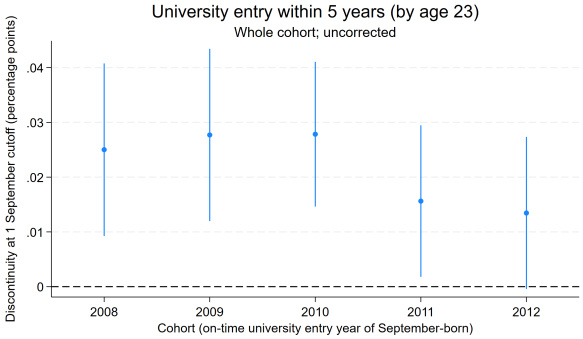
Figure 1: University Enrolment: Annual Discontinuities, 2008–2012



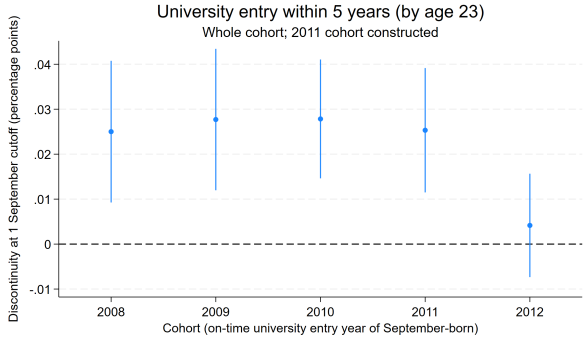
(a) On-time entry, original



(b) On-time entry, 2011 cohort constructed



(c) Entry within five years, original



(d) Entry within five years, 2011 cohort constructed

Note: Outcomes are defined for the whole cohort: on-time entry is entry in the academic year following the individual's 18th birthday, and entry within five years is entry by age 23. Panels (a) and (c) use observed outcomes, the primary specification; panels (b) and (d), a sensitivity analysis, use outcomes for the 2011 entry cohort constructed from the entry process of earlier cohorts (Section 5.6 and Appendix B). Each panel plots the regression discontinuity estimates for the 2008–2012 cutoffs within a 24-month window; formal estimates are in Table 2 and Appendix Table B.1. Data come from the 1998–2020 National Pupil Database and Higher Education Statistics Agency records.

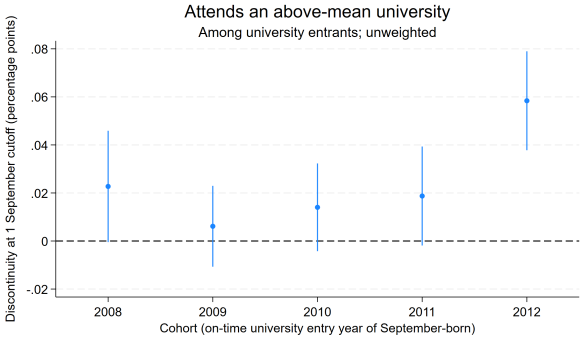
trated entirely below the mean, would by itself raise the above-mean share among entrants by about 1.2 points; adding the cohort-level increase of 1.23 points raises the implied conditional change to about 4.7 points, close to the estimated 4.41 (the calculation is approximate, because the estimation samples differ). Roughly a quarter of the conditional increase therefore reflects the smaller pool of entrants, and the remainder a modest increase in the share of the cohort attending selective institutions. Taken together, the point estimates for five-year entry and above-mean attendance imply that entry to below-mean institutions fell by about 2.3 points, while attendance at institutions between the mean and the top quartile was essentially unchanged.

Two qualifications apply. First, the split between selection and reallocation depends on the participation estimate. With the corrected five-year estimate of -2.21 points and the correspondingly smaller corrected cohort-level estimates (0.52 points above the mean, $p = 0.097$; 0.86 points in the top quartile, $p < 0.001$), selection accounts for a larger share (Appendix Table B.2). Second, reweighting the 2011 entrants raises the conditional estimates to 5.99 and 5.27 points, consistent with the 2011 entrant pool having been tilted toward students bound for selective institutions. In every specification, however, the conditional increases are several times the cohort-level ones, so changes in the composition of entrants cannot be read as improvements in the opportunities taken up by the eligible cohort as a whole.

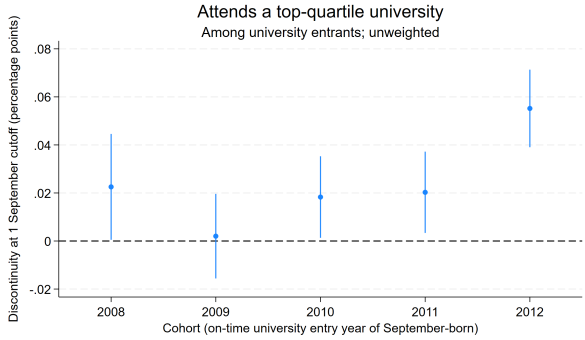
6.3 The AAB Exemption

The concurrent exemption of students with AAB or higher from student number controls freed institutions to recruit more high-attaining students in 2012–13, and could account for some of the shift toward selective institutions independently of the fee increase (Section 3.3). Excluding students who achieved AAB or higher (Appendix Table A.2; Figure 3 and Appendix Figure A.8), on-time entry fell by 3.33 points. Conditional on entry, the probability of attending an above-mean institution rose by 5.22 points and that of attending a top-quartile institution by 4.41 points (both $p < 0.001$), at least as much as in the full sample (4.41 and 3.74 points). The selectivity response

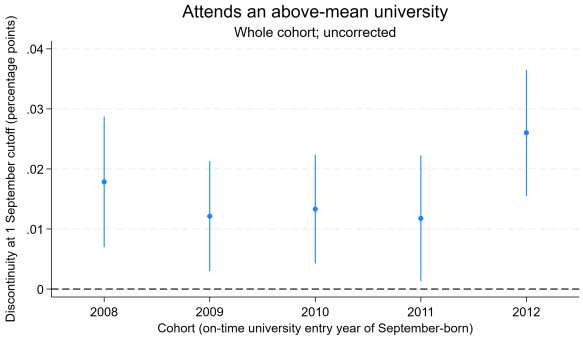
Figure 2: Selectivity of the Institution Attended: Annual Discontinuities



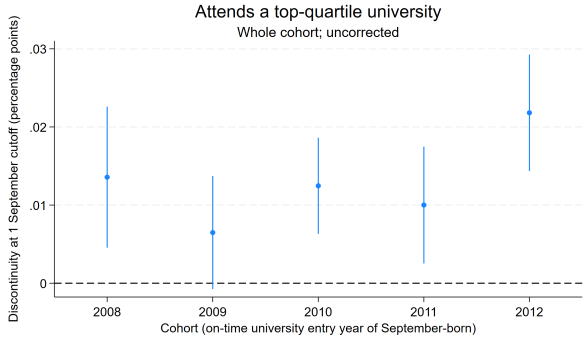
(a) Above-mean institution, among entrants



(b) Top-quartile institution, among entrants



(c) Above-mean institution, whole cohort



(d) Top-quartile institution, whole cohort

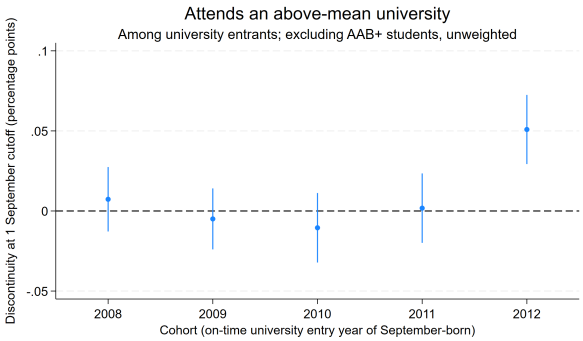
Note: Outcomes are indicators for attending an institution above the mean, or in the top quartile, of the selectivity distribution. Panels (a) and (b) are defined among university entrants; panels (c) and (d) are defined for the whole cohort and equal zero for individuals who do not enter university. All panels use observed, unweighted outcomes, the primary specification; corrected versions are in Appendix Figure B.1. Each panel plots the regression discontinuity estimates for the 2008–2012 cutoffs within a 24-month window; formal estimates are in Table 2. Data come from the 1998–2020 National Pupil Database and Higher Education Statistics Agency records.

is therefore not confined to students covered by the exemption, which is important evidence that the exemption is not its sole explanation. The top-10-percent estimate is about half as large as in the full sample (0.74 against 1.47 points), consistent with the exemption mattering more at the most selective institutions, where high-attaining entrants are concentrated by construction of the ranking. Excluding AAB+ students does not isolate the fee reform perfectly. The policy covered AAB "or equivalent" qualifications (Higher Education Funding Council for England 2011), whereas the exclusion uses A-level grades; A-level results were obtained after the reform was announced, so the exclusion selects on a potentially affected outcome; and the exemption and the accompanying reallocation of controlled places to lower-fee providers could have affected non-exempt students indirectly. Pooled estimates of whole-cohort selectivity excluding AAB+ students are not reported; panels (c) and (d) of Figure 3 show the annual discontinuities.

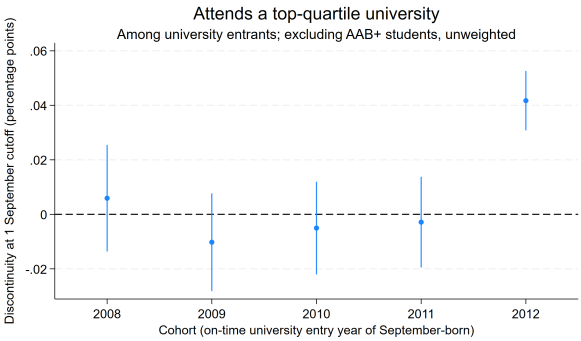
6.4 Subject Choice, Completion, and Progression

Appendix Table A.3 reports the remaining education outcomes, and Appendix Figures A.9 and A.10 show the annual discontinuities. Across the whole cohort, STEM participation fell by 0.70 percentage points ($p = 0.081$); among entrants, the STEM share fell by 0.96 points ($p = 0.326$). Neither estimate is significant at the 5 percent level. After correction, the cohort-level decline is larger and highly significant (1.53 points), while the conditional estimate remains insignificant, so any decline in STEM participation appears to operate through lower participation rather than through subject choice among entrants. With a STEM share among entrants of about one-third, a fall of 1.05 points in participation would by itself reduce cohort-level STEM participation by about 0.36 points, half the uncorrected estimate. Among entrants, the point estimates for degree completion (-1.26 points), withdrawal for poor academic performance (-1.27) and postgraduate study ($+0.51$) are not significant at the 5 percent level, with or without reweighting. Delayed entrants have less time to complete a degree or progress to postgraduate study before the end of the data, and the treated cohort delayed entry more;

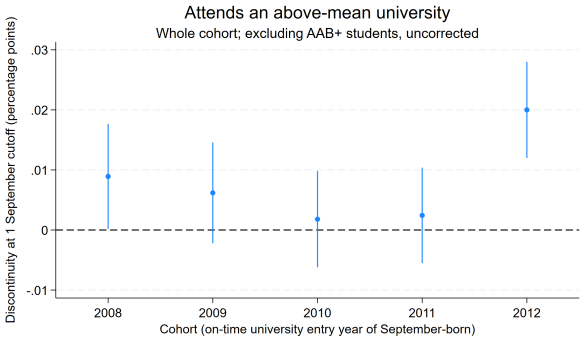
Figure 3: Excluding Students Covered by the AAB Exemption: Annual Discontinuities



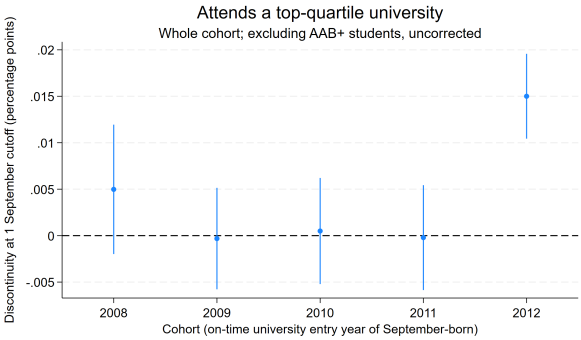
(a) Above-mean institution, conditional



(b) Top-quartile institution, conditional



(c) Above-mean institution, unconditional



(d) Top-quartile institution, unconditional

Note: The sample excludes students achieving AAB or higher at A-level. Panels (a) and (b) are defined among university entrants; panels (c) and (d) are defined for the whole cohort and equal zero for individuals who do not enter. All panels use observed, unweighted outcomes; the construction and reweighting of the 2011 cohort have not been applied. Data come from the 1998–2020 National Pupil Database and Higher Education Statistics Agency records.

postgraduate study is also affected by the introduction of postgraduate loans in 2016.

6.5 Early Labour-Market Outcomes

Table 3 reports the effects on employment, out-of-work benefit receipt and log annual earnings at each age from 22 to 25. For each outcome and age, it shows the pooled pre-reform discontinuity, the 2012 discontinuity, their difference, its standard error and its 95 percent confidence interval. The underlying discontinuities can be large, most clearly for earnings, which at age 22 are about 18 percent lower just above the cutoff than just below it, but they are similar in the 2012 and pre-reform cohorts, so the differences-in-discontinuities are small.

Employment. The reform had no statistically detectable effect on employment. The estimates are -0.13 percentage points at age 22, -0.46 at 23 and -0.45 at 24 (p between 0.34 and 0.78), and -0.41 at age 25. The confidence intervals exclude reductions in employment larger than about 1.1 points at age 22 and 1.4 points at ages 23 and 24, and increases larger than about 0.5 to 0.8 points. These bounds are informative about the cohort as a whole, but not about the individuals whose education the reform changed. The reform reduced the on-time entry of about 4 percent of the cohort. If its employment effects operated only through that change, a reduction of 1.4 points in cohort employment would correspond to a reduction of about 35 percentage points among those who did not enter on time. The calculation is only indicative, because the reform also changed the timing of entry and where students enrolled, but it shows that the estimates rule out large effects on the cohort, not economically meaningful effects on the minority whose education changed.

Out-of-work benefits. The estimates for out-of-work benefit receipt are $+0.66$ percentage points at age 22 ($p = 0.164$), $+0.24$ at 23, -0.19 at 24 and $+0.44$ at 25. None is statistically significant, they show no consistent pattern across ages, and their confidence intervals lie between

Table 3: The Effect of the Reform on Early Labour-Market Outcomes

	(1) Pre-reform $\widehat{\tau}_{2008\text{–}2011,k}$	(2) 2012 $\widehat{\tau}_{2012,k}$	(3) Difference $\widehat{\Delta}_k$	(4) 95% CI	(5) p
Panel A. Employed (percentage points)					
Age 22	-1.03	-1.16	-0.13	$[-1.07,\ 0.81]$	0.783
	(0.23)	(0.42)	(0.48)		
Age 23	0.29	-0.17	-0.46	$[-1.39,\ 0.48]$	0.337
	(0.22)	(0.42)	(0.48)		
Age 24	0.86	0.41	-0.45	$[-1.39,\ 0.48]$	0.343
	(0.22)	(0.43)	(0.48)		
Age 25	1.18	0.76	-0.41	$[-1.35,\ 0.53]$	0.390
	(0.22)	(0.43)	(0.48)		
Panel B. Receives out-of-work benefits (percentage points)					
Age 22	-1.32	-0.66	0.66	$[-0.27,\ 1.59]$	0.164
	(0.23)	(0.41)	(0.47)		
Age 23	0.01	0.25	0.24	$[-0.65,\ 1.12]$	0.600
	(0.22)	(0.39)	(0.45)		
Age 24	0.06	-0.13	-0.19	$[-1.04,\ 0.66]$	0.662
	(0.20)	(0.38)	(0.43)		
Age 25	-0.38	0.06	0.44	$[-0.41,\ 1.29]$	0.307
	(0.19)	(0.39)	(0.43)		
Panel C. Log annual earnings, 2015 prices, among those with earnings (log units)					
Age 22	-0.197	-0.215	-0.018	$[-0.048,\ 0.012]$	0.229
	(0.007)	(0.013)	(0.015)		
Age 23	*not available in the current output*				
Age 24	-0.069	-0.070	-0.001	$[-0.028,\ 0.026]$	0.925
	(0.006)	(0.012)	(0.014)		
Age 25	-0.039	-0.036	0.003	$[-0.024,\ 0.030]$	0.838
	(0.006)	(0.012)	(0.014)		

Notes: Age is $18 + k$, where outcomes are measured in the tax year k years after the on-time university entry year of individuals born just after each cutoff; individuals on both sides of a cutoff are observed in the same tax year. Column (1) is the pooled regression discontinuity estimate for the 2008–2011 cutoffs at that horizon, column (2) the estimate for the 2012 cutoff, and column (3) their difference (equation 5). Each is a local linear RDHonest estimate with a triangular kernel within a 12-month window. Eicker–Huber–White standard errors in parentheses; the standard error of the difference assumes zero covariance. Column (4) is $\widehat{\Delta}_k \pm 1.96\, SE(\widehat{\Delta}_k)$ and column (5) the conventional two-sided p-value; neither is bias-aware. Employment and out-of-work benefit receipt are defined for the whole cohort (986,275 observations at the pre-reform cutoffs and 242,865 at the 2012 cutoff) and reported in percentage points. Log annual earnings are in 2015 prices, defined for individuals with recorded earnings (pre-reform / 2012 samples: 740,780 / 192,760 at age 22; 773,940 / 196,530 at age 24; 775,485 / 195,870 at age 25), and reported in log units, where 0.01 is approximately 1 percent. The age-23 earnings estimate is not available in the current output. The age-25 observation for the 2012 cohort is in the 2019–20 tax year, which includes the start of the COVID-19 pandemic. Uncorrected specification; Appendix Table B.3 reports the 2011-adjusted estimates. Counts rounded to the nearest five. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$.

about -1.0 and $+1.6$ points. The data therefore provide no statistically detectable evidence that the fall in university entry increased reliance on out-of-work benefits in early adulthood. Employment and out-of-work benefit receipt measure different margins: individuals can be neither employed nor receiving out-of-work benefits, for example while in education or economically inactive, so the two need not move in opposite directions. The point estimates are nonetheless directionally consistent with each other, negative for employment at every age and positive for benefit receipt at three of four, although every estimate is small relative to its standard error, and estimates at different ages, which come from the same individuals, are not independent.

Earnings. Among individuals with recorded earnings, the estimated effects on log annual earnings are -0.018 at age 22 ($p = 0.229$), -0.001 at age 24 and $+0.003$ at age 25; the age-23 estimate is not available. The confidence intervals span roughly -4.8 to $+1.2$ percent at age 22 and about ± 3 percent at ages 24 and 25. The large negative underlying discontinuities, -0.197 at age 22 before the reform, reflect the education and experience differences described in Section 5.5, and the difference-in-discontinuities removes them: the 2012 discontinuities are almost identical. Because earnings are observed only for individuals with recorded earnings, these estimates combine any change in pay with any change in who has earnings. The small employment estimates suggest that the share in work did not change substantially, but they cannot rule out changes in its composition. With that qualification, there is no statistically detectable earnings response.

Industry. Appendix Table A.4 reports effects on employment in above-mean-paying and top-quartile-paying industries, defined for the whole cohort. The estimates at ages 22 to 24 are small and insignificant. At age 25, they are $+1.63$ points for above-mean-paying industries ($p = 0.008$) and $+1.20$ points for top-quartile-paying industries ($p = 0.038$), the only conventionally significant estimates among the labour-market outcomes, and the point estimates for both indicators rise with age. I do not give these results much weight. Neither age-25 estimate survives a Bonferroni

correction across the eight age-by-industry tests (threshold $0.05/8 \approx 0.006$), the paper reports many more labour-market comparisons than these eight, and the age-25 observation for the 2012 cohort falls in a tax year that includes the start of the pandemic. I treat them as exploratory rather than as evidence of beneficial sorting.

Age 25. The age-25 estimates are reported for completeness. The 2019–20 tax year includes the start of the COVID-19 pandemic and the first national lockdown in March 2020, near the end of the tax year. This affects only the 2012 cohort's age-25 observation, not the pre-reform benchmark, and could have shifted the experience-related discontinuity in 2012 for reasons unrelated to the reform; the direction of any such effect is not known. The employment, benefit and earnings estimates at age 25 are similar to those at earlier ages, so this caveat does not change the conclusions for those outcomes. It is more consequential for the industry estimates, which are significant only at that age.

Sensitivity to the 2011 adjustment. Reweighting the 2011 cohort by predicted on-time entry changes the employment and benefit estimates by at most 0.03 percentage points at any age, and the available adjusted earnings estimates (-0.015 at age 22; essentially zero at age 23) are similar to the uncorrected ones (Appendix Table B.3). The conclusions are therefore robust to this particular adjustment. That is not the same as robustness to all forms of selection or confounding: the adjustment targets selection into on-time university entry, not into employment or into having earnings (Section 5.5).

7 Discussion: Education Responses and Labour-Market Outcomes

The results separate several margins that aggregate statistics combine. *University entry decisions*: on-time entry fell by 4.0 percentage points. *Delayed entry*: around 70 percent of that decline was reversed within a year. *Selection into the attending sample*: participation within five years was between 1.0 and 2.2 points lower, depending on the treatment of the 2011 cohort, so entrants were a smaller and differently selected group. *Sorting between institutions*: among entrants, attendance at above-mean and top-quartile institutions rose by roughly 4 points, but across the cohort by only about 1 point. *Subject choice among entrants*: no detectable change. *Early labour-market outcomes*: no statistically detectable change in employment, out-of-work benefit receipt or earnings at ages 22 to 25. Changes measured among entrants cannot be read as changes for the underlying population, and a change in education does not by itself imply a detectable change in the labour market.

Why so little trace in early labour-market outcomes? Several explanations are consistent with the contrast between a large education response and small, imprecise labour-market estimates. None is separately identified here, and they are not mutually exclusive.

Dilution. The estimates are reduced-form effects on the whole cohort at the cutoff. The reform changed the on-time entry of about 4 percent of the cohort and, depending on the specification, the five-year participation of 1 to 2 percent. Even large effects on those individuals would be small in the cohort average: an effect confined to 2 percent of the cohort is diluted fifty-fold.

Timing. Students who delayed or forwent entry entered the labour market earlier and accumulated experience while their counterparts were studying. At ages 22 to 25, this experience can offset the earnings advantage of additional education, so early outcomes understate any longer-run difference.

Returns that emerge later. Graduate earnings premia in England grow with age: at 29, Belfield et al. (2018) estimate average returns to attending higher education of 26 percent for women and 6 percent for men, with men's returns still rising after 30. A cohort-level effect would be small at the ages observed here even if the forgone education had average returns.

Heterogeneous returns. The students deterred at the margin may have expected lower returns than the average entrant, consistent with evidence that returns are lower for students at the margin of enrolment (Carneiro et al. 2011). Evidence that academically marginal students admitted to college earn returns close to the average (Zimmerman 2014) cautions against assuming that the deterred students had low returns.

Substitution. Some of those who did not enter may have taken up other education or training, which the labour-market outcomes here do not separately identify.

Offsetting composition. The reform also changed where entrants studied. Greater attendance at selective institutions among those who entered could raise their later earnings, offsetting losses among those who did not.

Precision. The confidence intervals remain compatible with effects that would be economically meaningful for the individuals whose education changed.

The results therefore do not show that university education has no causal labour-market return. The paper estimates the effect of exposure to a tuition-fee reform on a cohort, not the effect of attending or completing university on those who did so. Nor do the results show that students deterred from university were better off entering employment, since their counterfactual trajectories are not observed. The strongest interpretation the evidence supports is that substantial educational responses to price need not translate into readily detectable short-run changes in aggregate labour-market outcomes.

Mechanisms for the selectivity response. Several mechanisms are consistent with the shift toward selective institutions among entrants, and the design does not distinguish between them.

The first is selection: a common increase in price deters the entrants for whom university is least attractive, and if these are disproportionately students who would have attended less selective institutions, the conditional share at selective institutions rises even if no individual changes destination. Section 6.2 shows that this mechanism accounts for roughly a quarter of the conditional estimate in the primary specification, and for more after correction. The second is greater attention to expected returns: a higher price may lead students who do enter to weigh more carefully where they enrol, raising relative demand for institutions perceived to offer higher returns. The increase in unconditional attendance at top-quartile institutions requires some mechanism beyond selection, because selection out of higher education cannot raise the share of the cohort attending a given institution. The third is supply and admissions. Number controls, the AAB exemption, and the reallocation of controlled places all changed in 2012, and institutions may have responded to the new fee regime in their recruitment. A reduction in delayed entry from the previous cohort could also have freed places at the institutions those students would have attended.

One feature of the English system makes the returns-based interpretation less straightforward than it may appear. Under income-contingent repayment with a write-off after 30 years, the expected cost of a higher fee rises with expected graduate earnings, because graduates with low earnings repay only part of their balance (Britton et al. 2017). For a student who responds only to expected repayments, the fee increase therefore raised the effective price of options with higher expected earnings by more than that of options with lower expected earnings. A shift toward selective institutions is more naturally explained by responses to the headline price or the size of the debt (Callender and Jackson 2005), by selection, or by changes on the supply side, than by a model in which students respond only to expected repayments. The results are consistent with students becoming more selective or returns-conscious, but they do not measure students' preferences or expectations directly.

Policy relevance. Income-contingent loans removed upfront costs, but they did not prevent a large immediate fall in university entry. In the four to seven years that followed, that fall was not accompanied by detectable deterioration in employment, earnings or benefit receipt for the cohort as a whole, which suggests that the short-run labour-market costs of the reform, averaged across the cohort, were small. It does not establish that the reform saved public money without cost: its fiscal and welfare consequences depend on the lifetime returns to the education forgone and to the reallocation across institutions, and on loan repayments and tax revenue that accrue well beyond the horizon observed here (Britton et al. 2020). Judging the reform by the characteristics of entrants would also mislead, since a rise in the share of entrants at selective institutions coincided with a fall in participation concentrated below the mean.

8 Robustness and Limitations

Inference. The main tests are conventional tests applied to RDHonest estimates with heteroskedasticity-robust standard errors. They are not bias-aware and do not capture cohort-level shocks (Section 5.4). The on-time entry estimate is large relative to its standard error; the cumulative-entry, selectivity and labour-market estimates are smaller and more dependent on the maintained assumptions.

The 2011 cohort. The persistence of the participation response depends on the correction for the final pre-reform cohort (Section 6.1), which rests on a model of the entry process extrapolated by one year, and whose standard errors do not reflect estimation of that model (Appendix B). The labour-market conclusions are insensitive to the corresponding adjustment, but that adjustment addresses selection into on-time entry only.

Labour-market identification. The labour-market estimates require that the experience-related discontinuity would have been stable across cohorts. The underlying discontinuities are

similar in 2012 and the pre-reform period, which is consistent with this assumption but does not establish it. Earnings are observed only for individuals with recorded earnings, and the age-25 observation for 2012 coincides with the start of the pandemic.

Multiple testing. The paper reports many outcome-by-horizon comparisons. Isolated significant estimates, such as the age-25 industry effects, should be read with this in mind.

Horizon and external validity. Labour-market outcomes are observed only to age 25, before graduate earnings premia have fully emerged, so the results speak to early careers rather than to lifetime earnings. The estimates are effects at the boundary between two cohorts, for individuals born around 1 September, of a single large reform in one system. Every member of the 2012 cohort faced the new fees, and there is no particular reason to expect individuals at the boundary to respond differently from the rest of the cohort, but relative age is correlated with attainment, and the response may vary with it.

9 Conclusion

This paper asks how students responded when the price of university in England nearly tripled under universal income-contingent loans, and whether that response left a trace in early labour-market outcomes. Comparing the discontinuity at the first treated school-entry cutoff with the pooled discontinuity at the four preceding cutoffs, I estimate the reform's effects on university entry, on where entrants enrolled, and on employment, out-of-work benefit receipt and earnings four to seven years after expected entry.

Three findings emerge. First, the reform substantially reduced immediate university participation: on-time entry fell by 4.0 percentage points. Most of this decline was offset by delayed entry. Participation within five years was 1.0 points lower in the primary specification, an estimate that is not significant at the 5 percent level, and 2.2 points lower, a highly significant

estimate, after correcting for the partial exposure of the final pre-reform cohort. The immediate response is robust; the evidence for a persistent contraction is suggestive without the correction and considerably stronger with it.

Second, the composition of university attendance changed. Among entrants, attendance at above-mean and top-quartile institutions rose by 4.4 and 3.7 points, and the shift is present among students not covered by the concurrent exemption of high attainers from number controls. Across the whole cohort, the increases were less than a third as large, so changes among entrants overstate changes in the opportunities taken up by the cohort.

Third, despite the large fall in on-time entry, there is no statistically detectable deterioration in employment, out-of-work benefit receipt or earnings at ages 22 to 25. The estimates rule out large effects on the cohort as a whole, but not economically meaningful effects on the minority whose education changed, and they say nothing about returns later in working life. The increases in employment in higher-paying industries at age 25 do not survive a correction for multiple testing and coincide with the onset of the pandemic.

The central lesson is that a large educational response to price need not produce readily detectable changes in aggregate labour-market outcomes in early adulthood. Whether it does so later, and what the reform implies for lifetime earnings, tax revenue and welfare, are questions for longer horizons than these data yet allow.

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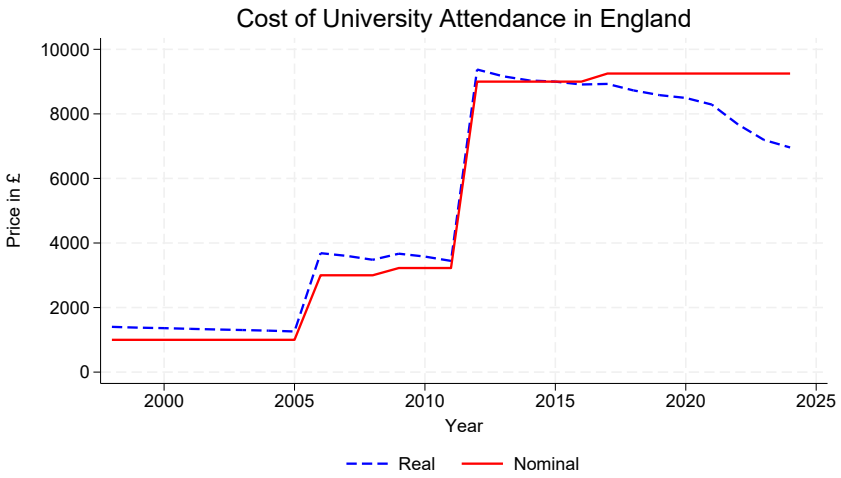
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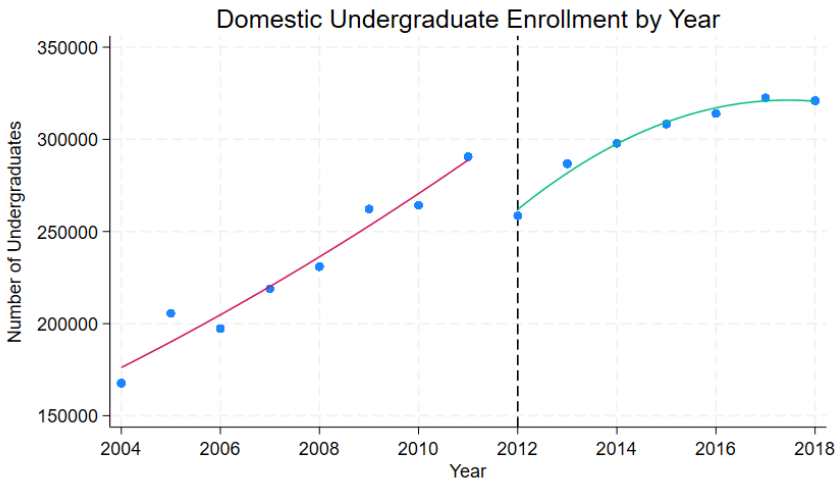
A Additional Figures and Tables

Figure A.1: Maximum Annual Tuition Fee, England



Note: Figure shows the maximum annual tuition fee chargeable to home undergraduate students in England, 1998–2024, in nominal and real (2024) prices. The 1998, 2006, and 2012 discontinuities correspond to the Teaching and Higher Education Act (1998), the Higher Education Act (2004), and the post-Browne reform respectively.

Figure A.2: Domestic First-Year Undergraduate Entrants by Year of Entry



Note: Figure shows the number of domestic first-year undergraduate entrants at English institutions by academic year of entry, 2004–2018. The dashed line marks the 2012–13 academic year, the first cohort subject to the £9,000 fee cap. Fitted trends are shown separately for the pre- and post-reform periods.

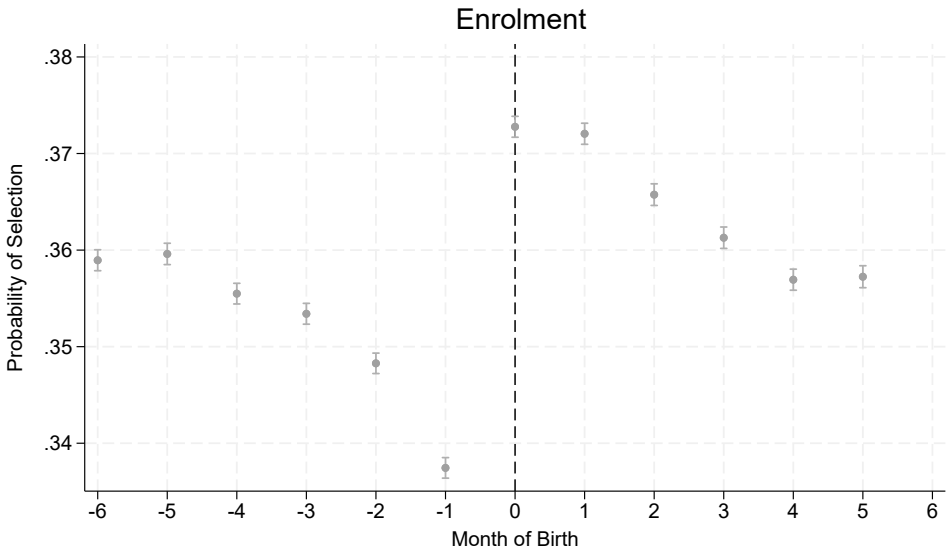


Figure A.3: Probability of university enrolment by month of birth, centred at 0 for September-born students, illustrating the discontinuity around the academic year cutoff.

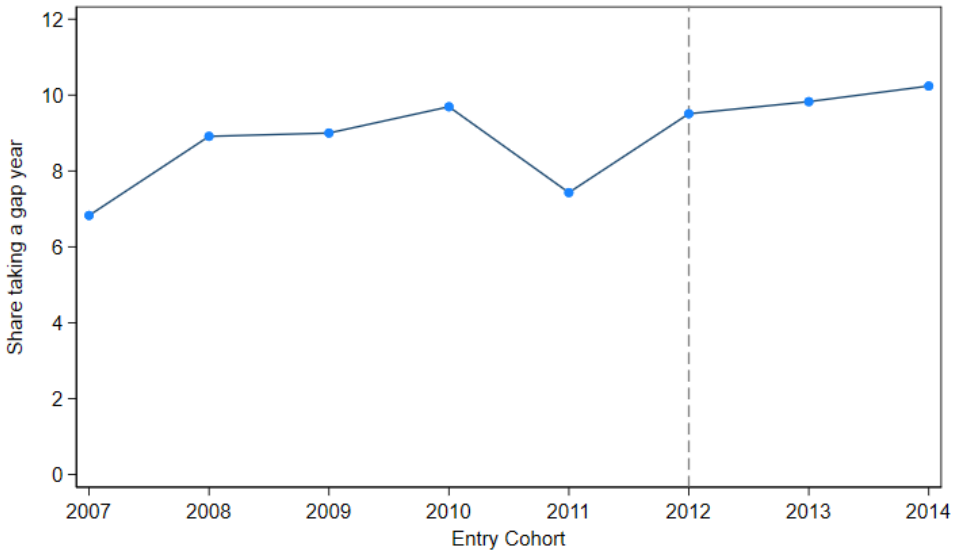


Figure A.4: Share of Students Delaying University Entry by One Year

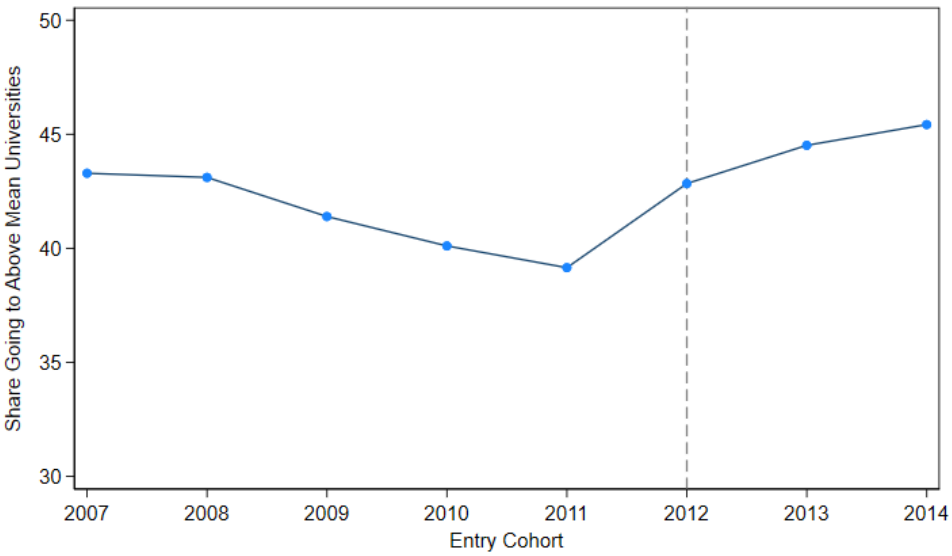


Figure A.5: Share of Entrants Attending Selective Institutions Over Time

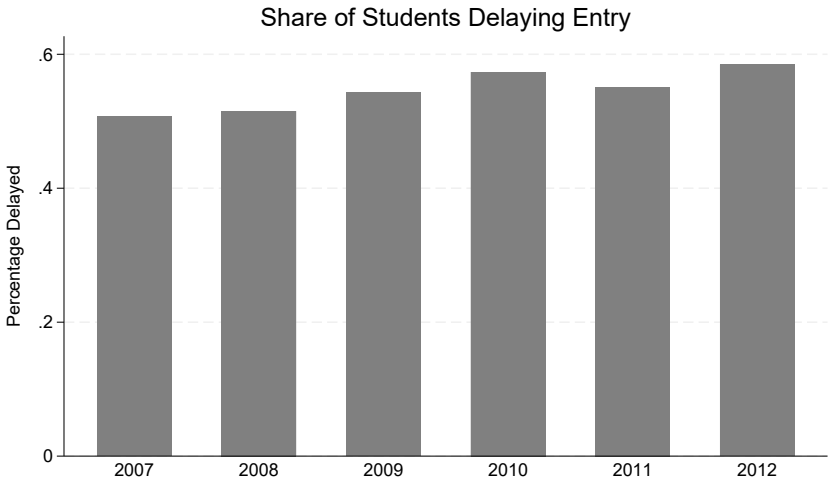


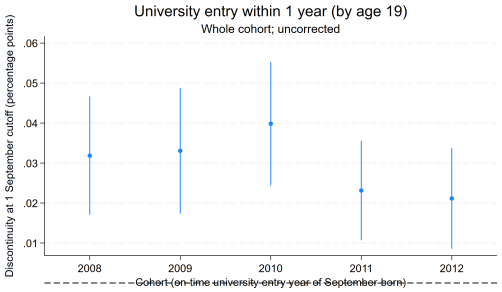
Figure A.6: Share of students who delay school entry by one year, by cohort.

Table A.1: Covariate Balance at the 1 September Cutoff, by Cohort

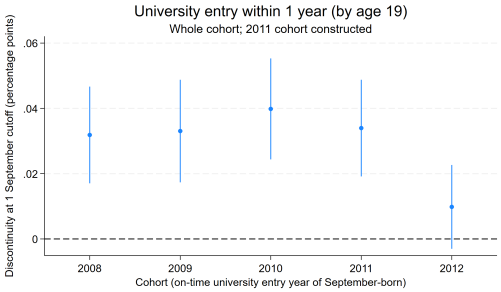
	(1) 2008	(2) 2009	(3) 2010	(4) 2011	(5) 2012
Female	-0.4	-0.3	-0.7	-0.5	-1.0
	$[-1.5,\ 0.6]$	$[-1.7,\ 1.1]$	$[-2.1,\ 0.7]$	$[-2.0,\ 1.1]$	$[-2.5,\ 0.6]$
N	337,105†	252,295	249,650	244,150	242,865
Eligible for FSM	-2.9^{*}	-0.8	-1.1^{*}	-0.7	-0.2
	$[-4.1,\ -1.7]$	$[-2.0,\ 0.3]$	$[-2.0,\ -0.2]$	$[-1.7,\ 0.3]$	$[-1.3,\ 0.9]$
N	240,180	252,295	249,650	244,150	242,865
Low-participation area	-0.2	0.7	0.3	-1.5^{*}	0.8
	$[-1.4,\ 1.1]$	$[-0.6,\ 1.9]$	$[-1.1,\ 1.7]$	$[-2.9,\ -0.1]$	$[-0.4,\ 2.0]$
N	240,180	252,295	249,650	244,150	338,325†

Notes: Each cell reports the regression discontinuity estimate at the cutoff indexed by the column year, in percentage points, with the reported 95 percent confidence interval in brackets. * indicates that the reported interval excludes zero. Counts rounded to the nearest five.

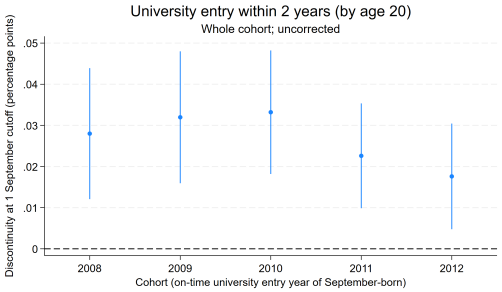
Figure A.7: Cumulative University Entry at Intermediate Horizons: Annual Discontinuities



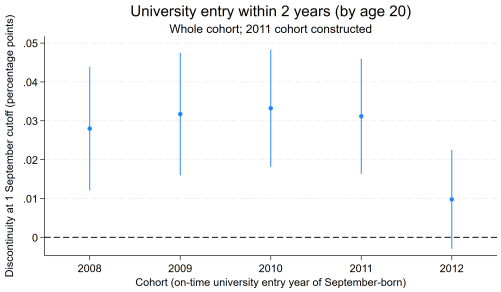
(a) Within 1 year, original



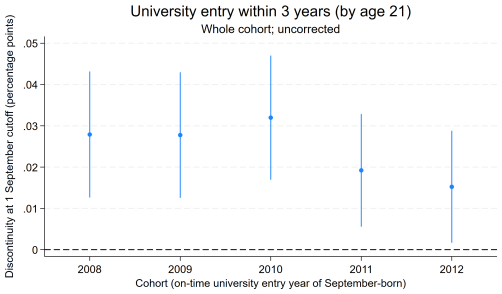
(b) Within 1 year, constructed



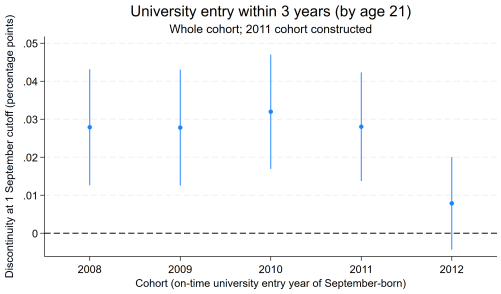
(c) Within 2 years, original



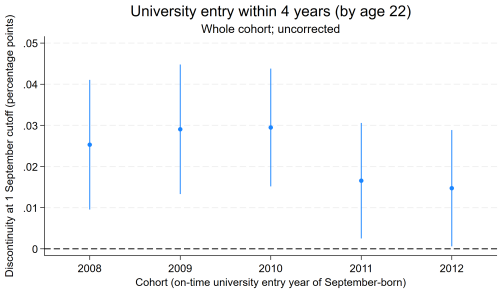
(d) Within 2 years, constructed



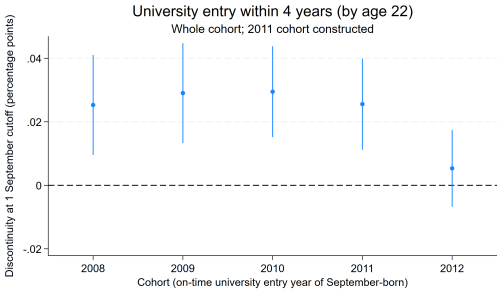
(e) Within 3 years, original



(f) Within 3 years, constructed



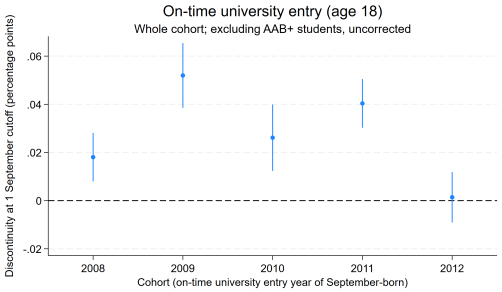
(g) Within 4 years, original



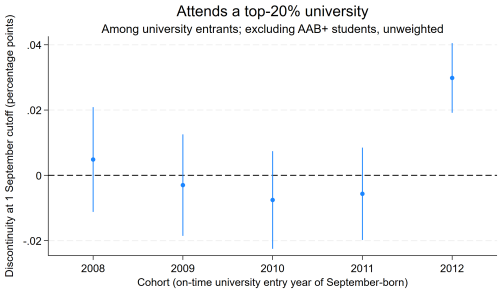
(h) Within 4 years, constructed

Note: Outcomes are indicators for entering university within one to four years of the on-time date, defined for the whole cohort. Left panels use observed outcomes; right panels use outcomes for the 2011 entry cohort constructed from the entry process of earlier cohorts.

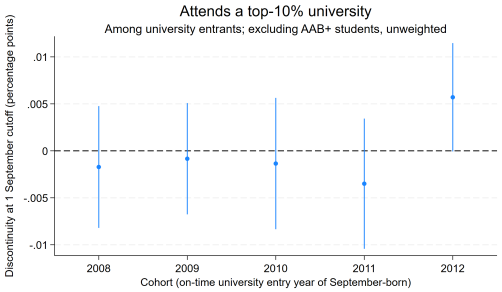
Figure A.8: Excluding AAB+ Students, Additional Outcomes: Annual Discontinuities



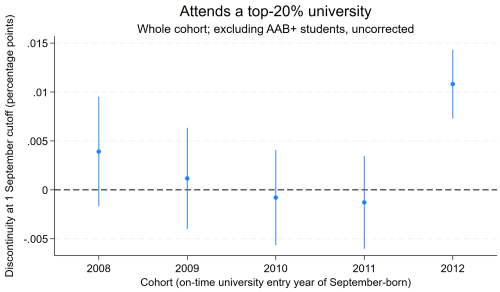
(a) On-time entry



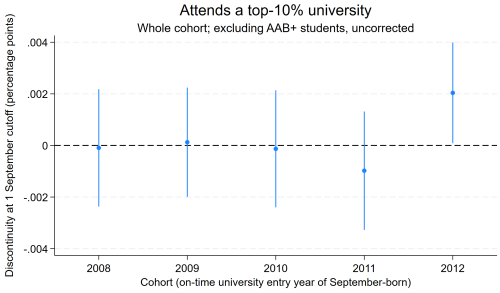
(b) Top 20%, conditional



(c) Top 10%, conditional



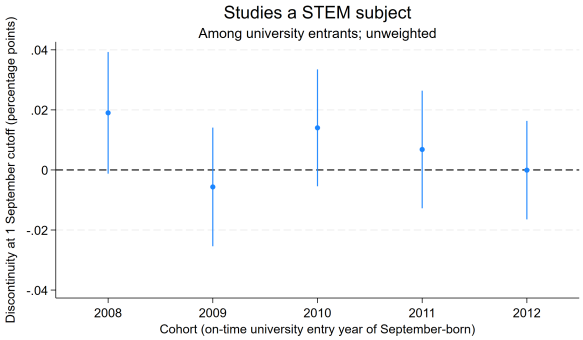
(d) Top 20%, whole cohort



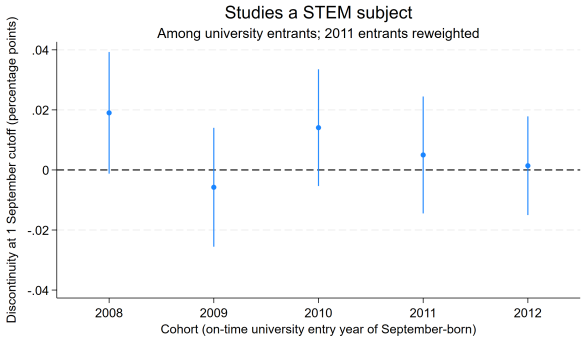
(e) Top 10%, whole cohort

Note: The sample excludes students achieving AAB or higher at A-level. All panels use observed, unweighted outcomes. Pooled estimates for the whole-cohort outcomes are not reported.

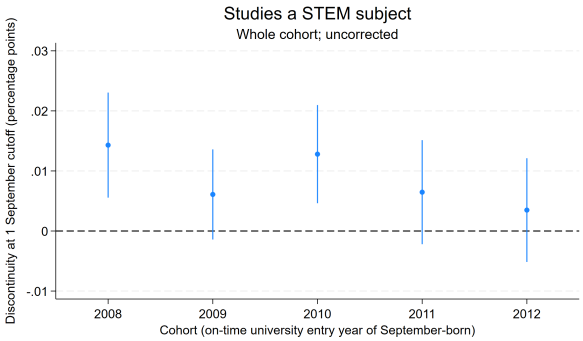
Figure A.9: STEM Participation: Annual Discontinuities



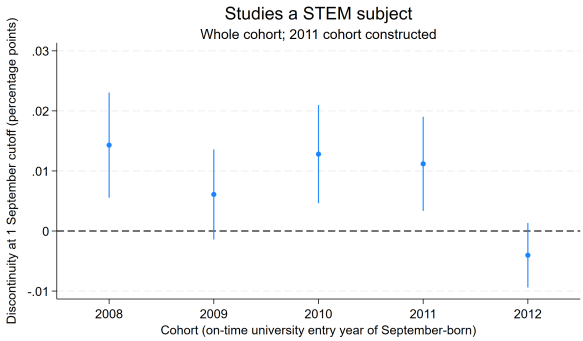
(a) STEM among entrants, unweighted



(b) STEM among entrants, reweighted



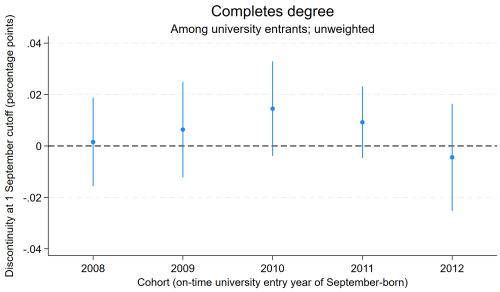
(c) STEM, whole cohort, original



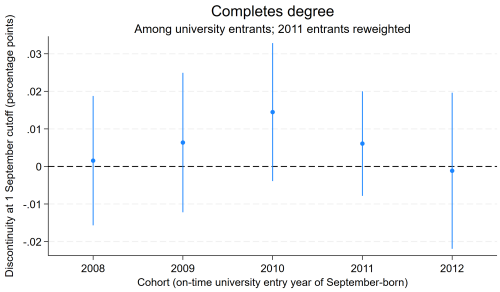
(d) STEM, whole cohort, constructed

Note: Panels (a) and (b) are defined among university entrants; panel (b) reweights the 2011 entrants. Panels (c) and (d) are defined for the whole cohort; panel (d) constructs the 2011 cohort's outcomes.

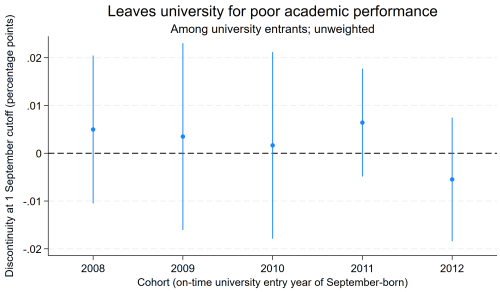
Figure A.10: Completion, Withdrawal, and Postgraduate Study: Annual Discontinuities



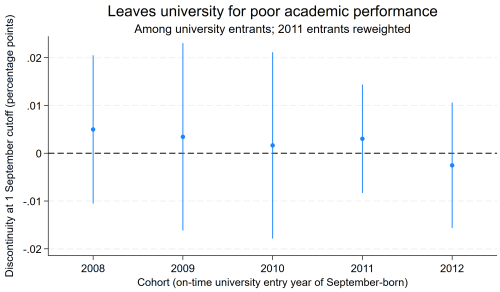
(a) Degree completion, unweighted



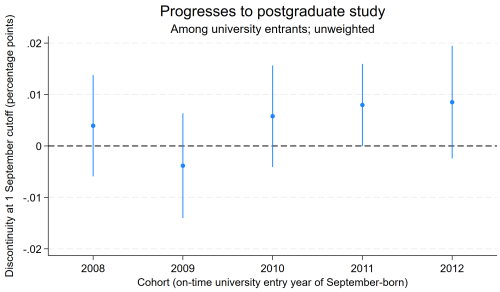
(b) Degree completion, reweighted



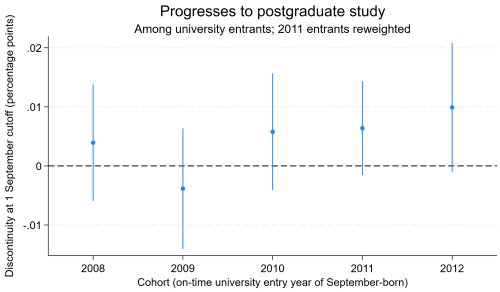
(c) Withdrawal (poor academic performance), unweighted



(d) Withdrawal (poor academic performance), reweighted



(e) Postgraduate study, unweighted



(f) Postgraduate study, reweighted

Note: All outcomes are defined among university entrants; right panels reweight the 2011 entrants.

Table A.2: Excluding Students Covered by the AAB Exemption

	(1) Full sample	(2) Excluding AAB+
On-time entry	-4.02^{***}	-3.33^{***}
	[<0.001]	[<0.001]
Conditional on university entry		
Above-mean institution	4.41^{***}	5.22^{***}
	[<0.001]	[<0.001]
Top-quartile institution	3.74^{***}	4.41^{***}
	[<0.001]	[<0.001]
Top 20% institution	2.50^{***}	3.12^{***}
	[0.002]	[<0.001]
Top 10% institution	1.47^{***}	0.74^{**}
	[0.003]	[0.012]

Notes: Each cell reports the pooled difference-in-discontinuities estimate in percentage points (equation 2), estimated as described in Table B.1. All estimates are uncorrected and unweighted; the AAB+ exclusion has not yet been combined with the construction or reweighting of the 2011 cohort. Column (2) excludes students achieving AAB or higher at A-level. Sample sizes (pre-reform / 2012): on-time entry 986,270 / 242,865 in column (1) and 927,160 / 227,260 in column (2); conditional outcomes 345,055 / 87,005 and 299,750 / 75,265. Pooled estimates of whole-cohort selectivity excluding AAB+ students are not reported. P-values in brackets are from the conventional two-sided test and are not bias-aware. Counts rounded to the nearest five. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.3: Subject Choice, Completion, and Progression

	(1) Uncorrected / unweighted	(2) Corrected / reweighted	(3) N pre-reform	(4) N 2012
Panel A. Whole cohort (column 2: 2011 cohort constructed)				
STEM subject	-0.70^{*}	-1.53^{***}	986,275	242,865
	[0.081]	[<0.001]		
Panel B. Conditional on university entry (column 2: 2011 entrants reweighted)				
STEM subject	-0.96	-0.77	349,390	88,035
	[0.326]	[0.428]		
Postgraduate study	0.51	0.68	349,390	88,035
	[0.328]	[0.186]		
Degree completion	-1.26	-0.91	347,160	87,200
	[0.133]	[0.285]		
Withdrawal for poor academic performance	-1.27^{*}	-0.90	316,885	79,580
	[0.069]	[0.204]		

Notes: Each cell reports the pooled difference-in-discontinuities estimate in percentage points (equation 2), estimated as described in Table B.1. Withdrawal for poor academic performance is constructed from the HESA reason for ending a course and is not a general measure of non-completion. P-values in brackets are from the conventional two-sided test; they are not bias-aware and do not account for estimation of the entry models. Counts rounded to the nearest five. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A.4: Employment in Higher-Paying Industries

	(1) Pre-reform $\widehat{\tau}_{2008\text{–}2011,k}$	(2) 2012 $\widehat{\tau}_{2012,k}$	(3) Difference $\widehat{\Delta}_k$	(4) 95% CI	(5) p
Panel A. Employed in an above-mean-paying industry (percentage points)					
Age 22	-0.62	-0.95	-0.33	$[-1.52,\ 0.87]$	0.591
	(0.27)	(0.54)	(0.61)		
Age 23	-1.16	-0.85	0.31	$[-0.89,\ 1.51]$	0.611
	(0.27)	(0.55)	(0.61)		
Age 24	-1.08	-0.21	0.87	$[-0.33,\ 2.07]$	0.156
	(0.27)	(0.55)	(0.61)		
Age 25	-1.00	0.64	1.63^{***}	$[0.43,\ 2.84]$	0.008
	(0.27)	(0.55)	(0.61)		
Panel B. Employed in a top-quartile-paying industry (percentage points)					
Age 22	0.44	0.50	0.06	$[-1.05,\ 1.16]$	0.920
	(0.26)	(0.50)	(0.56)		
Age 23	-0.58	-0.02	0.56	$[-0.55,\ 1.67]$	0.321
	(0.26)	(0.50)	(0.56)		
Age 24	-0.94	-0.02	0.93	$[-0.19,\ 2.05]$	0.104
	(0.26)	(0.51)	(0.57)		
Age 25	-1.32	-0.12	1.20^{**}	$[0.07,\ 2.34]$	0.038
	(0.26)	(0.52)	(0.58)		

Notes: As for Table 3. The outcomes are indicators for being employed in an industry whose mean pay is above the mean, or in the top quartile, of industries ranked by mean pay in the 2015 tax year. Their estimation samples are the whole cohort (986,275 at the pre-reform cutoffs and 242,865 at the 2012 cutoff), as for employment. A Bonferroni correction across the eight age-by-industry tests requires $p < 0.05/8 = 0.00625$ for significance at the 5 percent level, which no estimate meets. The age-25 observation for the 2012 cohort is in the 2019–20 tax year, which includes the start of the COVID-19 pandemic. These estimates are exploratory. Uncorrected specification. Counts rounded to the nearest five. $^{*}\ p < 0.10$, $^{**}\ p < 0.05$, $^{***}\ p < 0.01$.

B The 2011 Cohort Correction

This appendix describes the correction for the partial exposure of the final pre-reform cohort summarised in Section 5.6, and reports the corrected estimates. The correction is applied to the whole 2011 entry cohort. It takes two forms for education outcomes, depending on whether the outcome is defined for the whole cohort or only for university entrants, and a third for labour-market outcomes.

Construction for unconditional outcomes. I estimate a logistic regression of each dependent variable on individual characteristics using the 2008–2010 cohorts. The covariates are month of birth, a female indicator, a LOPAR indicator, FSM eligibility, the number of GCSEs achieved at each grade (A*, A, B, C and so forth), school-LSOA fixed effects, and a linear cohort trend. Using the estimated model, I predict the outcomes that every member of the 2011 cohort would have had under the entry process of earlier cohorts, extending the cohort trend by one year, and use these predictions in place of the 2011 cohort's observed unconditional outcomes. The corrected estimate compares the 2012 cohort with a 2011 cohort whose outcomes reflect the earlier entry process. It targets θ, the effect of the new fee regime relative to the old one for individuals at the 2012 cutoff, under two assumptions. First, absent the reform, the 2011 cohort's outcomes given its observed characteristics would have followed the process estimated on earlier cohorts, with the trend extended by one year. This rules out departures from trend in the 2011 cohort that are unrelated to the reform, which the correction would wrongly remove, and relies on extrapolating a trend estimated over years that include the recession. Second, because the corrected outcomes enter discontinuity estimates in which the relative-age profile is the object of interest, the predictions must reproduce that profile. The model includes month of birth, so they reproduce the profile estimated on the 2008–2010 cohorts; the correction therefore assumes that the 2011 cohort's counterfactual profile matches theirs, the same stability assumption on which the design rests.

Reweighting for conditional outcomes. Outcomes such as the selectivity of the institution attended are defined only for university entrants. For these outcomes the problem is different: the reform's effect on the 2011 cohort changes who is observed entering, so the 2011 entrants may be selected differently from earlier entrants. I therefore reweight the members of the 2011 cohort who entered university in 2011 by

$$w_i \;=\; \frac{\widehat{P}_{\text{pre}}(E_i = 1 \mid X_i)}{\widehat{P}_{2011}(E_i = 1 \mid X_i)}, \tag{6}$$

where E_i indicates on-time university entry and X_i are the covariates above. The numerator is the predicted probability of on-time entry from the model estimated on cohorts before 2011; the denominator is the corresponding prediction from a model with the same covariates estimated on the 2011 cohort (in which the cohort trend is not identified). To see what the weights do, let $f_{2011}(X)$ denote the distribution of characteristics in the 2011 cohort. Among 2011 entrants, characteristics are distributed in proportion to $P_{2011}(E = 1 \mid X) f_{2011}(X)$; had the same cohort been selected by the earlier process, they would have been distributed in proportion to $P_{\text{pre}}(E = 1 \mid X) f_{2011}(X)$. The weight is the ratio of the two, up to a constant, so the weighted 2011 entrants have the composition that the earlier selection process would have produced from the 2011 cohort, in the spirit of DiNardo et al. (1996). Entrants whose characteristics made them more likely to enter in 2011 than earlier entry patterns predict receive less weight; those whose entry became less likely receive more.

Reweighting addresses selection on observed characteristics. It recovers the counterfactual conditional outcome under the assumption that, given X_i, entrants' outcomes, such as where they enrolled and what they studied, are the same as they would have been under the earlier process. It does not address changes in where entrants with given characteristics enrolled, for instance students accepting less selective places to secure entry before the fee increase. Nor does it identify the full set of counterfactual outcomes for individuals who would have entered

under the earlier process but did not: their characteristics are represented only through observed entrants with similar characteristics, which requires every type that would have entered under the earlier process to have a positive probability of entering in 2011.

Construction and reweighting are different operations. Unconditional outcomes are constructed; conditional outcomes are reweighted. Construction replaces the 2011 cohort's outcomes with what the earlier process predicts for the whole cohort. Reweighting leaves observed outcomes unchanged and alters only the relative weight of observed entrants. The two rest on different assumptions and target different objects, and the paper does not use them interchangeably.

Inference for corrected estimates. The standard errors for corrected estimates treat the fitted entry models as if their coefficients were known. Constructed outcomes are predicted probabilities, which vary less across individuals than realised binary outcomes, which is why the constructed estimates have smaller standard errors than the uncorrected ones; if the coefficients were known, this gain in precision would be legitimate. They are estimated, however, and the resulting prediction errors are shared: an error in the trend coefficient shifts the prediction for every member of the 2011 cohort, and an error in a school-LSOA fixed effect shifts it for every student in that school neighbourhood. Shared errors of this kind do not average out across the 2011 cohort, and the reported p-values do not reflect them. Nor do they reflect uncertainty about whether the 2011 cohort would have continued the trend of earlier cohorts, which is a modelling assumption rather than sampling error.[3]

Remaining contamination. Only the 2011 cohort is corrected, but every earlier cohort also contains individuals who delayed entry long enough to face the new fees: members of the 2010

[3] As an informal check, I evaluate each constructed estimate against the standard error of the corresponding uncorrected estimate, which is computed from realised outcomes. Every constructed estimate that is significant at the 5 percent level remains so. The check does not capture estimation error in the entry model.

cohort who delayed by two years and, at horizons of three to five years, members of earlier cohorts who delayed for longer. Exposure is therefore present in every pre-reform cohort and grows toward 2011. Its effect on the estimates depends on where each cohort sits in the design. Let δ_k denote the reduction in a cumulative outcome that this exposure causes in cohort k, and suppose for the moment that it is the same for the cohort's older and younger members. Because cohort k supplies the treatment side of cutoff k and the comparison side of cutoff $k+1$, it lowers τ_k by δ_k and raises τ_{k+1} by δ_k, so each annual discontinuity changes by $\delta_{k-1} - \delta_k$, the difference in exposure between adjacent cohorts. With exposure rising toward 2011, the annual pre-reform discontinuities for cumulative outcomes should therefore drift downward In the pooled estimate, however, these differences should difference out. With equal weights, the pooled pre-reform discontinuity changes by $(\delta_{2007} - \delta_{2011})/4$, and once the 2011 cohort is corrected, by $\delta_{2007}/4$ alone: the exposure of the 2008–2010 cohorts cancels, because each of those cohorts appears once on each side of a pre-reform cutoff.[4] What remains is (i) the 2007 term, from members of the 2007 cohort who entered five years late, which is small and pushes the estimate away from zero; (ii) differences in exposure between a cohort's older and younger members, which attenuate the estimate if exposure is concentrated among older members, who sit above the earlier cutoff, and work in the opposite direction if it is concentrated among younger members; and (iii) departures from equal weighting in the pooled fit, which are small because the cohorts are of similar size. Remaining contamination is therefore a plausible but uncertain source of attenuation, and likely to be second-order relative to the contamination of the 2011 cohort, which enters with coefficient $-(1 + \omega_{2011})$ and is unambiguously attenuating. At the one-year horizon only the 2011 cohort could have entered under the new fees, so the corrected one-year estimate is not affected by remaining contamination from earlier cohorts. I do not interpret any corrected estimate as a bound.

[4]For example, if exposure lowered five-year entry by 0.1, 0.2 and 0.4 percentage points in the 2008, 2009 and 2010 cohorts and by 0.05 points in the 2007 cohort (illustrative values), then with the 2011 cohort corrected the discontinuities at the 2008–2011 cutoffs would change by -0.05, -0.1, -0.2 and $+0.4$ points. These sum to 0.05, so the pooled pre-reform discontinuity would change by about 0.01 points.

Adjustment for labour-market outcomes. For labour-market outcomes, the 2011 cohort is reweighted using predicted probabilities of on-time university entry from the models above, applied to those who entered on time and, as one minus that probability, to those who did not, so that the cohort's mix of on-time entrants and non-entrants resembles what the entry process of earlier cohorts would have produced. The adjustment shares the assumptions of the reweighting for conditional outcomes, and has two further limitations. The characteristics that predict on-time university entry need not be those that determine employment, earnings or industry, and the adjustment does not address selection into employment or into having recorded earnings, either of which could itself respond to the reform. It is therefore a sensitivity analysis, not a correction that identifies the counterfactual distribution of labour-market outcomes. Table B.3 compares the uncorrected and adjusted estimates. Tables B.1 and B.2 report the corrected education estimates, and Figures B.1 and B.2 show the corresponding annual discontinuities.

Table B.1: University Entry: Uncorrected and Corrected Estimates

	(1) Uncorrected	(2) 2011 cohort constructed
On-time entry	-4.02^{***}	-4.57^{***}
	[<0.001]	[<0.001]
Entered within 1 year	-1.14^{**}	-2.51^{***}
	[0.040]	[<0.001]
Entered within 2 years	-1.15^{**}	-2.15^{***}
	[0.043]	[<0.001]
Entered within 3 years	-1.16^{**}	-2.11^{***}
	[0.045]	[<0.001]
Entered within 4 years	-1.08^{*}	-2.22^{***}
	[0.062]	[<0.001]
Entered within 5 years	-1.05^{*}	-2.21^{***}
	[0.072]	[<0.001]
N, pre-reform cutoffs (2008–2011)	986,275	986,275
N, 2012 cutoff	242,865	242,865

Notes: Each cell reports $\widehat{\Delta} = \widehat{\tau}_{2012} - \widehat{\tau}_{2008\text{–}2011}$ in percentage points (equation 2), where each discontinuity is a local linear RDHonest estimate within a 12-month window (six birth months on each side of the 1 September cutoff) and the pre-reform discontinuity is estimated on the pooled observations from the 2008–2011 cutoffs. Outcomes are defined for the whole cohort. Column (2) replaces the outcomes of the 2011 entry cohort with outcomes constructed from the entry process of earlier cohorts (Section 5.6). P-values in brackets are from a conventional two-sided test using $SE(\widehat{\Delta}) = [SE(\widehat{\tau}_{2012})^2 + SE(\widehat{\tau}_{2008\text{–}2011})^2]^{1/2}$; they are not bias-aware, and those in column (2) do not account for estimation of the entry model. The pre-reform sample is 986,270 for on-time and five-year entry. Counts rounded to the nearest five. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.2: Institution Selectivity: Uncorrected and Corrected Estimates

Panel A. Conditional on university entry		
	(1)	(2)
	Unweighted	2011 entrants reweighted
Above-mean institution	4.41^{***}	5.99^{***}
	[<0.001]	[<0.001]
Top-quartile institution	3.74^{***}	5.27^{***}
	[<0.001]	[<0.001]
Top 20% institution	2.50^{***}	3.87^{***}
	[0.002]	[<0.001]
Top 10% institution	1.47^{***}	2.06^{***}
	[0.003]	[<0.001]
N, pre-reform / 2012	345,055 / 87,005	345,055 / 87,005
Panel B. Whole cohort (unconditional)		
	(1)	(2)
	Uncorrected	2011 cohort constructed
Above-mean institution	1.23^{***}	0.52^{*}
	[0.004]	[0.097]
Top-quartile institution	1.14^{***}	0.86^{***}
	[<0.001]	[<0.001]
Top 20% institution	0.75^{**}	0.71^{***}
	[0.011]	[0.001]
Top 10% institution	0.48^{***}	0.42^{***}
	[0.007]	[<0.001]
N, pre-reform / 2012	986,275 / 242,865	986,275 / 242,865

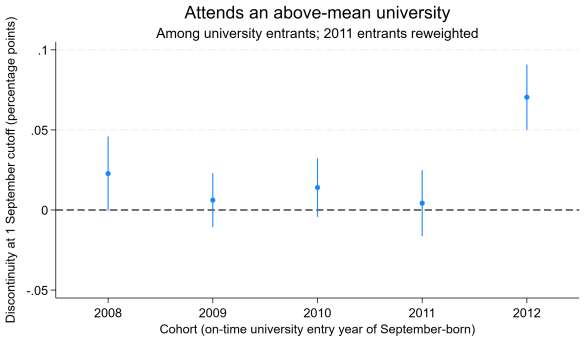
Notes: Each cell reports the pooled difference-in-discontinuities estimate in percentage points (equation 2), estimated as described in Table B.1. Institutions are ranked by the mean A-level points of their entrants in a pre-reform reference year. Panel A outcomes are defined among university entrants; column (2) reweights the 2011 entrants by equation (6). Panel B outcomes are defined for the whole cohort and equal zero for individuals who do not enter; column (2) replaces the outcomes of the 2011 cohort with constructed outcomes. P-values in brackets are from the conventional two-sided test; they are not bias-aware and do not account for estimation of the entry models. Counts rounded to the nearest five. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table B.3: Labour-Market Estimates: Uncorrected and 2011-Adjusted

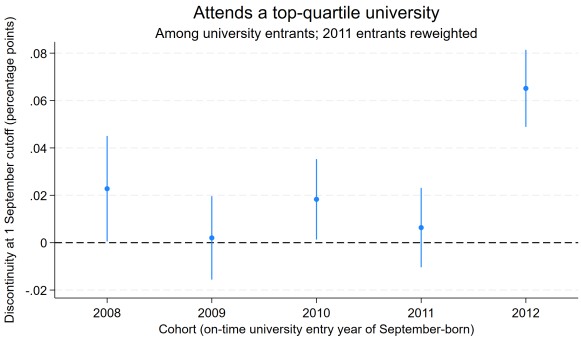
	(1) Uncorrected		(2) 2011-adjusted	
	$\widehat{\Delta}_k$ (SE)	p	$\widehat{\Delta}_k$ (SE)	p
Employed (percentage points)				
Age 22	-0.13 (0.48)	0.783	-0.13 (0.48)	0.792
Age 23	-0.46 (0.48)	0.337	-0.46 (0.48)	0.338
Age 24	-0.45 (0.48)	0.343	-0.48 (0.48)	0.313
Age 25	-0.41 (0.48)	0.390	-0.40 (0.48)	0.407
Receives out-of-work benefits (percentage points)				
Age 22	0.66 (0.47)	0.164	0.66 (0.48)	0.163
Age 23	0.24 (0.45)	0.600	0.23 (0.45)	0.617
Age 24	-0.19 (0.43)	0.662	-0.18 (0.43)	0.685
Age 25	0.44 (0.43)	0.307	0.47 (0.43)	0.279
Log annual earnings, among those with earnings (log units)				
Age 22	-0.018 (0.015)	0.229	-0.015 (0.015)	0.337
Age 23	—		0.000 (0.014)	0.980
Age 24	-0.001 (0.014)	0.925	—	
Age 25	0.003 (0.014)	0.838	—	
Employed in an above-mean-paying industry (percentage points)				
Age 22	-0.33 (0.61)	0.591	-0.21 (0.61)	0.725
Age 23	0.31 (0.61)	0.611	0.41 (0.61)	0.502
Age 24	0.87 (0.61)	0.156	0.97 (0.61)	0.116
Age 25	1.63^{***} (0.61)	0.008	1.67^{***} (0.62)	0.007
Employed in a top-quartile-paying industry (percentage points)				
Age 22	0.06 (0.56)	0.920	0.09 (0.57)	0.874
Age 23	0.56 (0.56)	0.321	0.61 (0.57)	0.281
Age 24	0.93 (0.57)	0.104	1.02^{*} (0.57)	0.075
Age 25	1.20^{**} (0.58)	0.038	1.25^{**} (0.58)	0.031

Notes: Each cell reports the difference-in-discontinuities estimate $\widehat{\Delta}_k$ (equation 5), with its standard error in parentheses, and the conventional p-value. Column (1) repeats Tables 3 and A.4; column (2) reweights the 2011 cohort using predicted probabilities of on-time university entry (Appendix B). Units, samples and definitions as in Table 3; the adjusted earnings samples are 740,780 / 192,760 at age 22 and 762,010 / 195,590 at age 23. "—" indicates an estimate not available in the current output. The standard errors in column (2) do not account for estimation of the weights. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

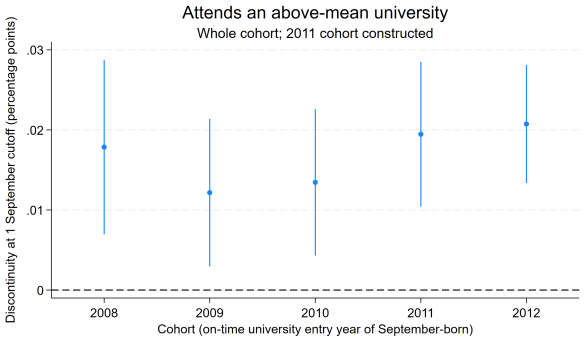
Figure B.1: Selectivity of the Institution Attended, 2011 Correction: Annual Discontinuities



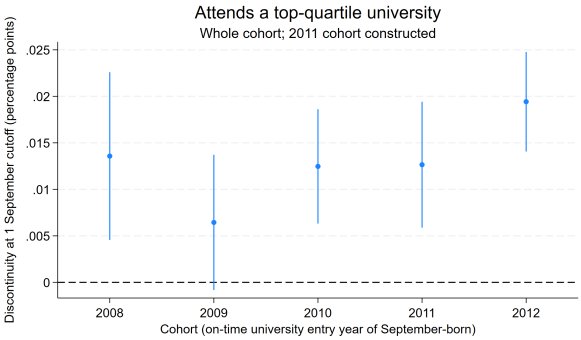
(a) Above-mean institution, among entrants, reweighted



(b) Top-quartile institution, among entrants, reweighted



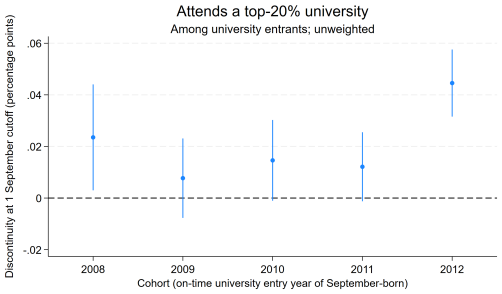
(c) Above-mean institution, whole cohort, constructed



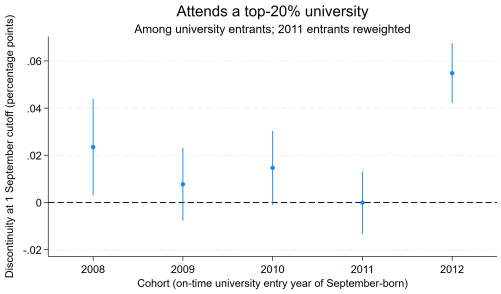
(d) Top-quartile institution, whole cohort, constructed

Note: Panels (a) and (b) are defined among university entrants and reweight the 2011 entrants (equation 6); panels (c) and (d) are defined for the whole cohort and use constructed outcomes for the 2011 cohort. Uncorrected versions are in Figure 2. Formal estimates are in Table B.2.

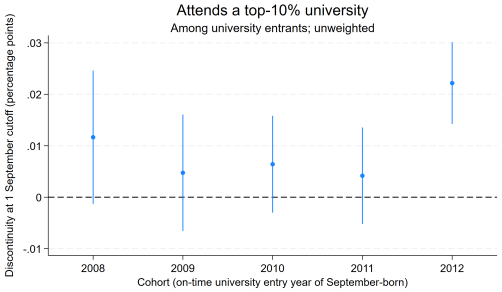
Figure B.2: Selectivity at Higher Thresholds: Annual Discontinuities



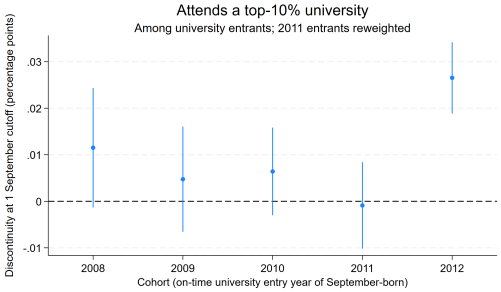
(a) Top 20%, conditional, unweighted



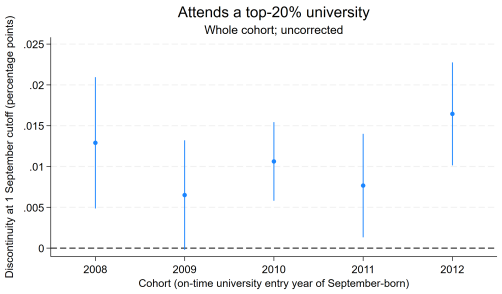
(b) Top 20%, conditional, reweighted



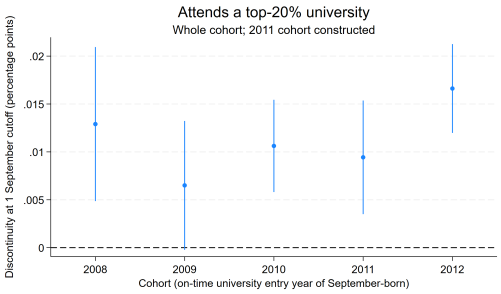
(c) Top 10%, conditional, unweighted



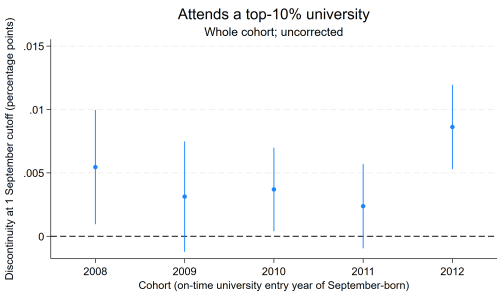
(d) Top 10%, conditional, reweighted



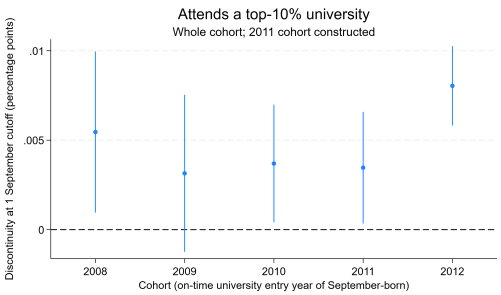
(e) Top 20%, whole cohort, original



(f) Top 20%, whole cohort, constructed



(g) Top 10%, whole cohort, original



(h) Top 10%, whole cohort, constructed

Note: Panels (a)–(d) are defined among university entrants, with the 2011 entrants reweighted in (b) and (d). Panels (e)–(h) are defined for the whole cohort, with the 2011 cohort's outcomes constructed in (f) and (h).